1. General Physics

YOUR NOTES
⬇

CONTENTS

1.1 Length & Time
1.1.1 Measurement
1.2 Motion
1.2.1 Speed & Acceleration
1.2.2 Distance-Time Graphs
1.2.3 Velocity-Time Graphs
1.2.4 Freefall
1.3 Mass & Weight
1.3.1 Mass & Weight
1.4 Density
1.4.1 Density
1.5 Forces
1.5.1 Change of Shape
1.5.2 Balanced Forces
1.5.3 Unbalanced Forces
1.5.4 Circular Motion
1.5.5 Turning Effect
1.5.6 Conditions For Equilibrium
1.5.7 Centre of Mass
1.5.8 Scalars & Vectors
1.6 Momentum
1.6.1 Momentum
1.7 Energy, Work & Power
1.7.1 Energy
1.7.2 KE & GPE
1.7.3 Energy Resources
1.7.4 Efficiency
1.7.5 Work
1.7.6 Power
1.8 Pressure
1.8.1 Pressure

1. General Physics

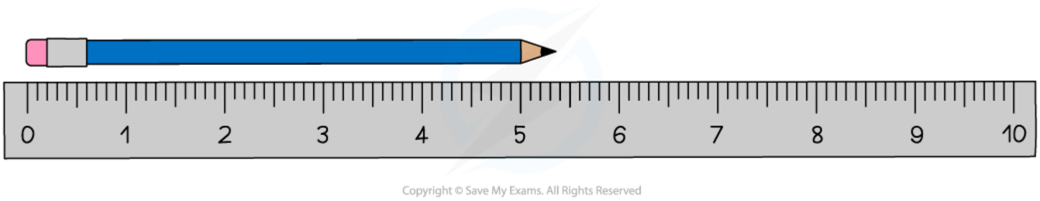
YOUR NOTES
⬇

1.1 LENGTH & TIME

1.1.1 MEASUREMENT

Distance & Volume

- Rulers can be used to measure small distances of a few cm. They are able to measure to the nearest mm

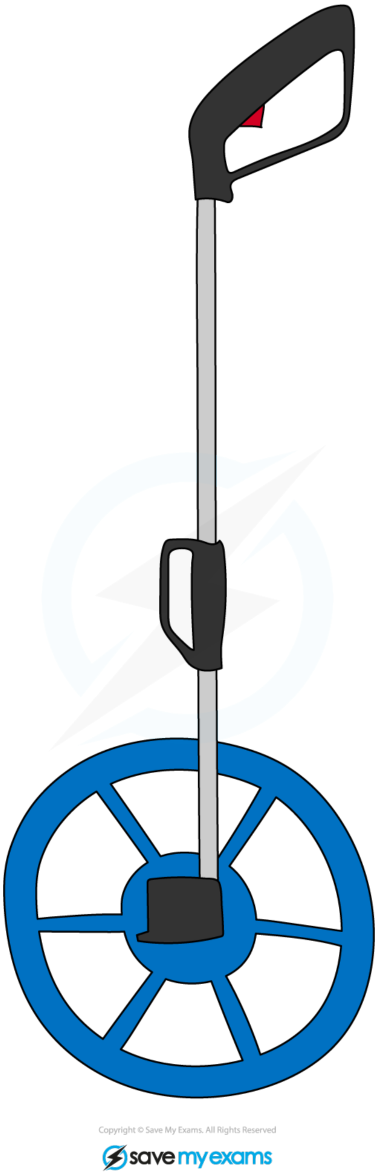


A ruler can measure small distances to the nearest mm

- When measuring larger distances (of a few metres) a tape measure is more appropriate or, when measuring even larger distances, a trundle wheel

1. General Physics

YOUR NOTES
⬇

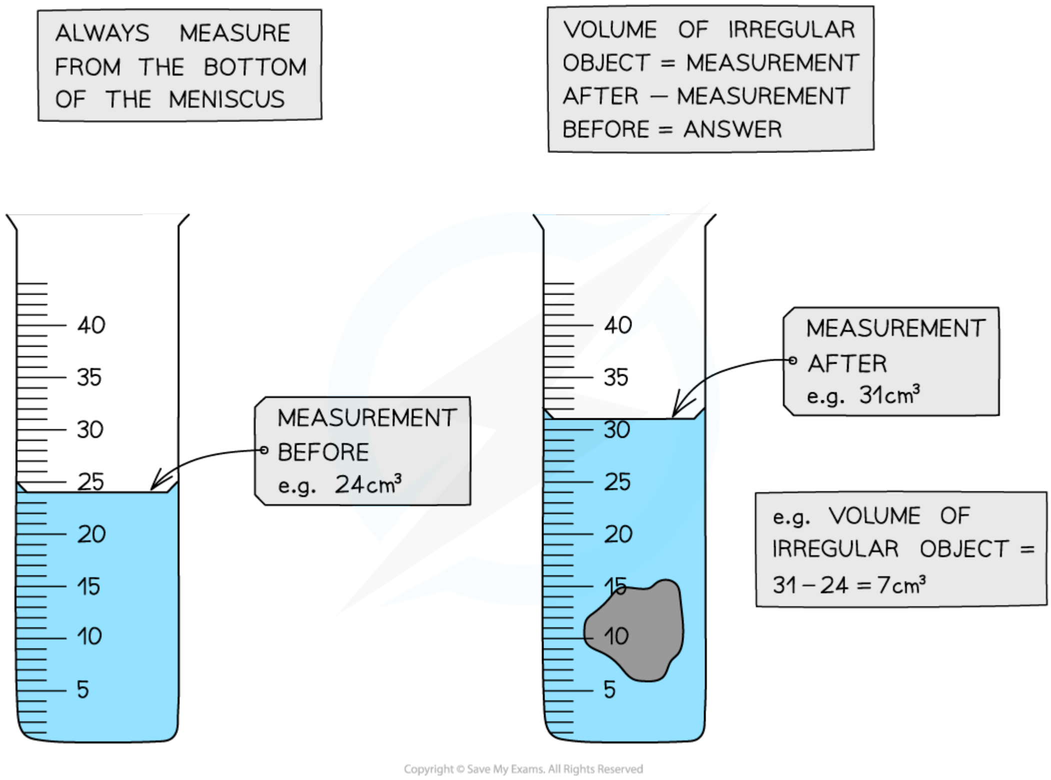


Trundle wheels can be used to measure large distances

1. General Physics

YOUR NOTES
⬇

- Measuring cylinders can be used to measure the volume of liquids or, by measuring the change in volume, the volume of an irregular shape



Measuring cylinders can be used to determine the volume of a liquid or an irregular shaped solid

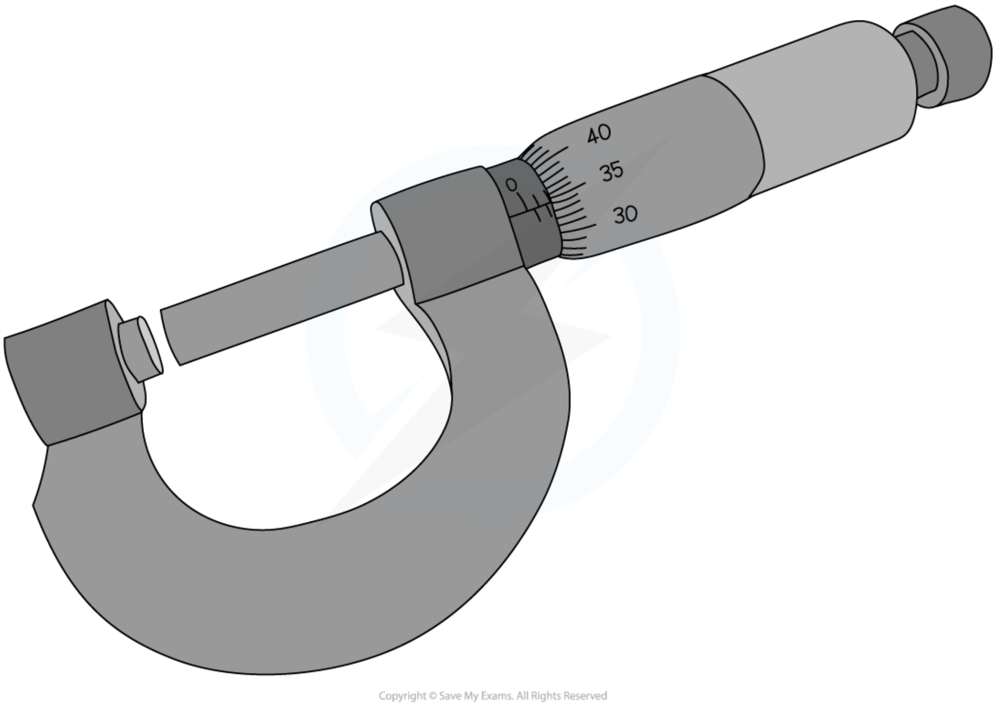
1. General Physics

YOUR NOTES

Extended Only

Micrometer Screw Gauge

- When measuring very small distances (less than a centimetre) a micrometer is the most appropriate instrument



Micrometers can be used to measure very small distances

- Micrometers can measure distances to the nearest 1/100th of a mm

1. General Physics

YOUR NOTES
⬇

Time

- Stop-clocks and stopwatches can be used to measure time intervals
- An important factor when measuring time intervals is human reaction time. This can have a significant impact upon measurements when the measurements involved are very short (less than a second)

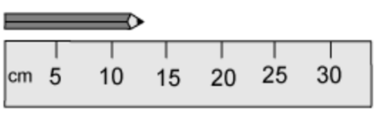
Multiple Readings

- Suppose you have to measure the thickness of a sheet of paper. The thing that you are trying to measure is so small that it would be very difficult to get an accurate answer
- If, however, you measure the thickness of 100 sheets of paper you can do so much more accurately. Dividing your answer by 100 will then give an accurate figure for the thickness of one sheet
- This process of taking a reading of a large number of values and then dividing by the number, is a good way of getting accurate values for small figures, including (for example) the time period of a pendulum – measure the time taken for 10 swings and then divide that time by 10

Exam Question: Easy

The image below shows an enlarged drawing of the end of a 30cm ruler

It is being used to measure the length of a pencil.



What is the length of the pencil?

A 16 cm

B 15 cm

C 13 cm

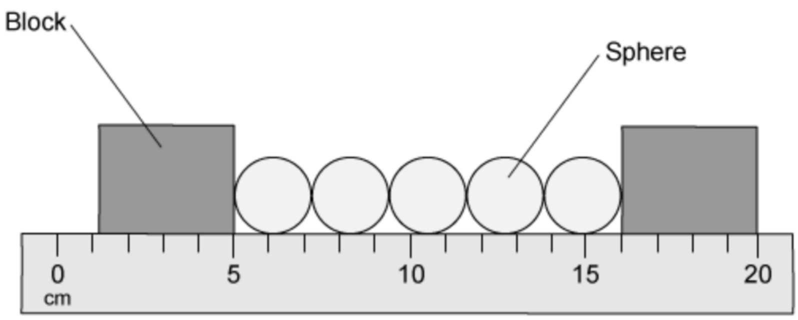
D 11 cm

1. General Physics

YOUR NOTES

Exam Question: Medium

Five identical spheres are placed between two blocks in order to measure their diameter.



What is the diameter of a single sphere?

A 2.2 cm

B 2.0 cm

C 2.1 cm

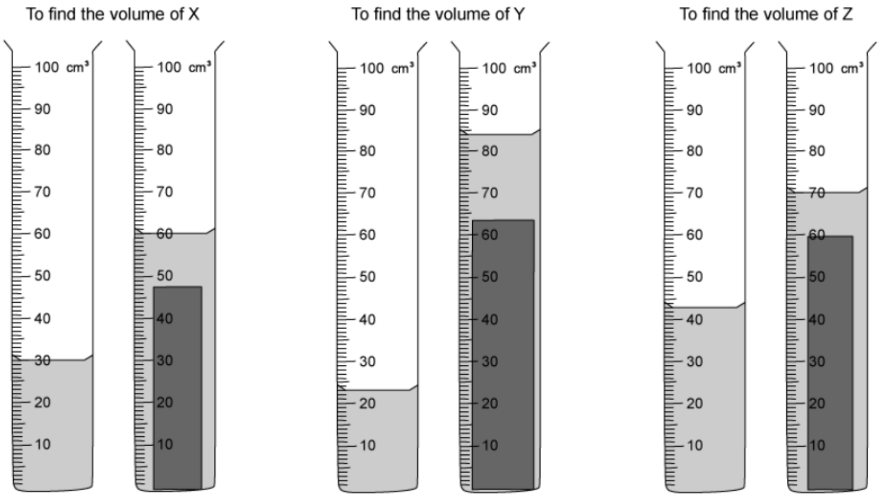
D 2.3 cm

1. General Physics

YOUR NOTES

Exam Question: Hard

Three blocks are placed into three measuring cylinders. These are shown below.



Which row in the table shows the blocks in order of increasing volume?

	Smallest volume	→	Largest volume
A	X	Y	Z
B	Y	X	Z
C	Z	Y	X
D	Z	X	Y

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

YOUR NOTES
⬇

1.2 MOTION

1.2.1 SPEED & ACCELERATION

Speed

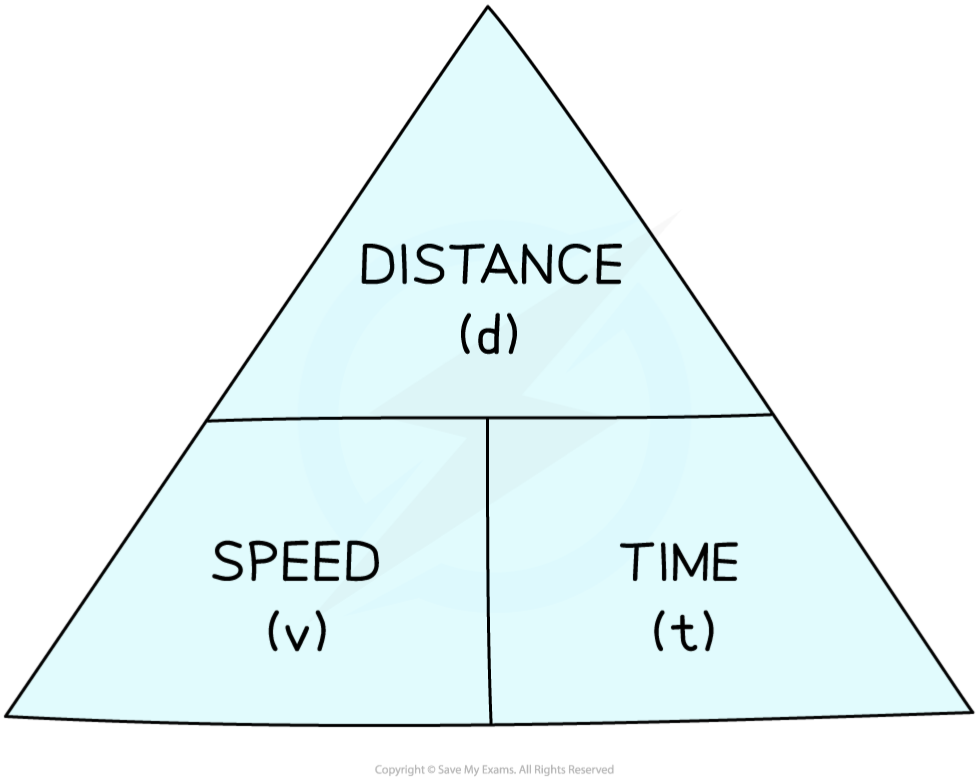
- Speed (measured in metres per second) is the distance moved by an object each second
- The average speed of an object is given by the equation:

$$\text{AVERAGE SPEED} = \frac{\text{DISTANCE MOVED}}{\text{TIME TAKEN}}$$

1. General Physics

YOUR NOTES ⬇

- You can rearrange the equation with the help of the formula triangle:



Use the formula triangle to help you rearrange the equation

Exam Tip

• Use the units of speed (**metres per second – distance divided by time**) to help you remember the formula
• The equation is for **average speed**, but the speed at a specific moment might be higher or lower

1. General Physics

YOUR NOTES

Extended Only

Velocity

- Velocity is a similar quantity to speed, but includes a direction (the direction of travel) as well as its value (its magnitude)
- Two objects can have equal speeds but might have opposite velocities (if they are travelling in opposite directions)

The two cars have the same speed but opposite velocities, because they are travelling in opposite directions

1. General Physics

YOUR NOTES
⬇

Extended Only

Acceleration

- **Acceleration is the rate of change of velocity**: In other words, how much the velocity of an object changes by every second
- Acceleration is given by the equation:

$$\text{ACCELERATION} = \frac{\text{CHANGE IN VELOCITY}}{\text{TIME TAKEN}}$$

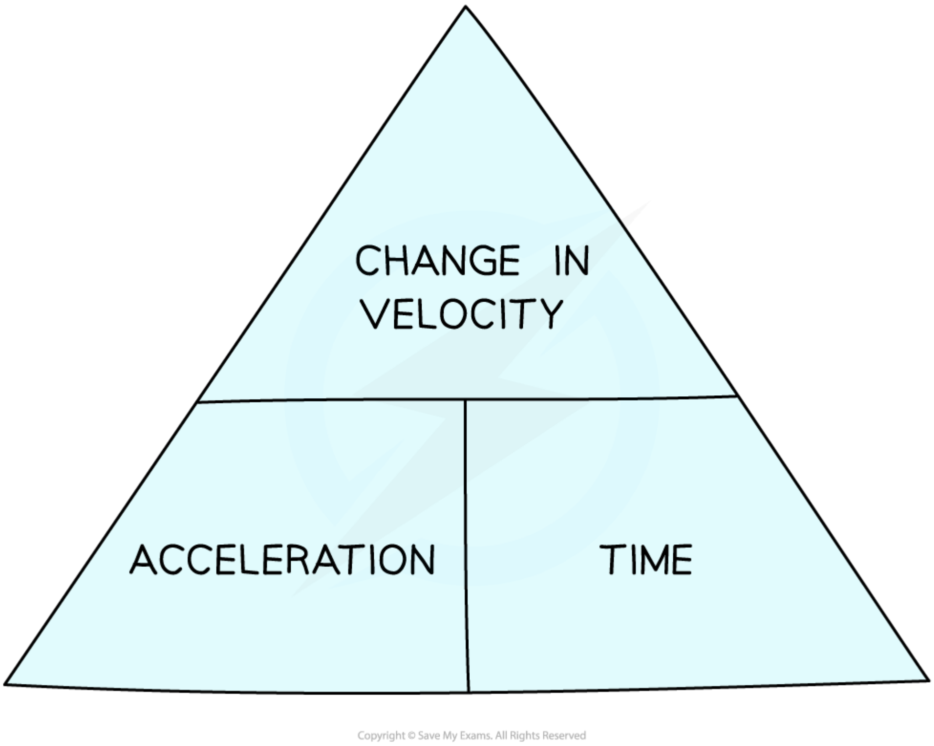
$$a = \frac{(v-u)}{t}$$

(Where ***u* is the initial velocity** of an object and ***v* is its final velocity**)

- You can rearrange this equation with the help of the formula triangle:

1. General Physics

YOUR NOTES
⬇



Use the formula triangle to help you rearrange the equation

- The units of acceleration are *m/s*^{2}, which mean the same thing as *m/s/s* – the change in velocity (in *m/s*) **every second**

Exam Tip

Marks are often available for giving the correct unit, even if your answer is incorrect. You must, however, give an answer (even if it's just a guess): giving a unit without an answer will not gain you any marks.

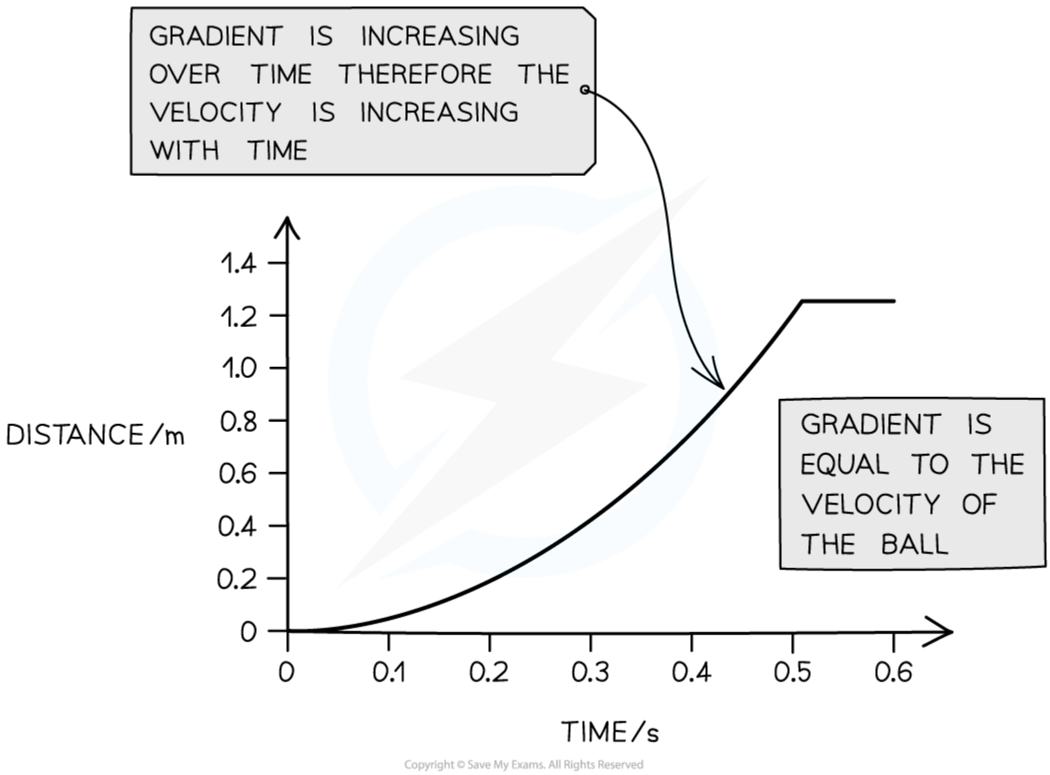
1. General Physics

YOUR NOTES

1.2.2 DISTANCE-TIME GRAPHS

Distance-Time Graphs: Basics

- A distance-time graph shows how the distance of an object (from a point) varies over time:



Graphs showing how the distances travelled by three objects vary over time

- A **horizontal** line means **stationary**
- A **straight** line means **constant speed**
- If the **gradient increases** the object is **speeding up** (accelerating)
- If the **gradient decreases** the object is **slowing down** (decelerating)
- If the line is **going down**, the object is moving **backwards**

1. General Physics

YOUR NOTES
⬇

Extended Only

Calculating Speed

- The **speed** of an object is given by the **gradien**t of the line

$$\text{SPEED} = \text{GRADIENT} = \frac{\text{RISE}}{\text{RUN}}$$

Exam Tip

When you come across any graph, look carefully at what is plotted on each axis and think for a while about what the graph is showing you. Distance-time graphs are also known as position-time graphs or displacement-time graphs. Don't be fooled by these different names: they describe the same kind of things.

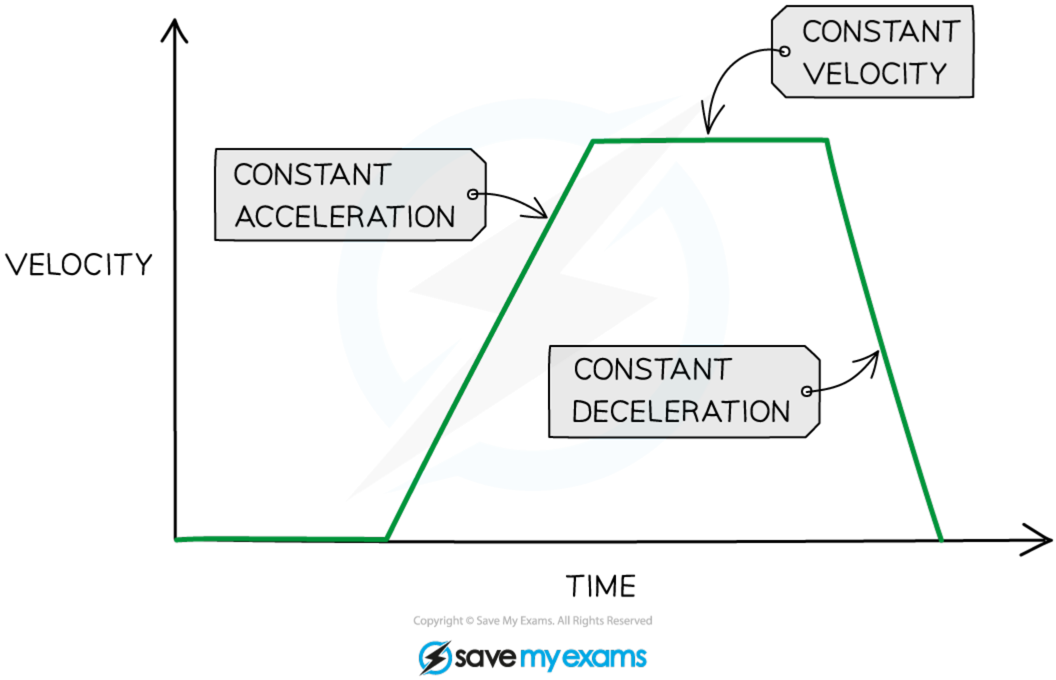
1. General Physics

YOUR NOTES
⬇

1.2.3 VELOCITY-TIME GRAPHS

Velocity-Time Graphs: Basics

- A Velocity-time graph shows how the velocity (or speed) of an object changes over time



Graph showing how the velocity (speed) of an object changes over time

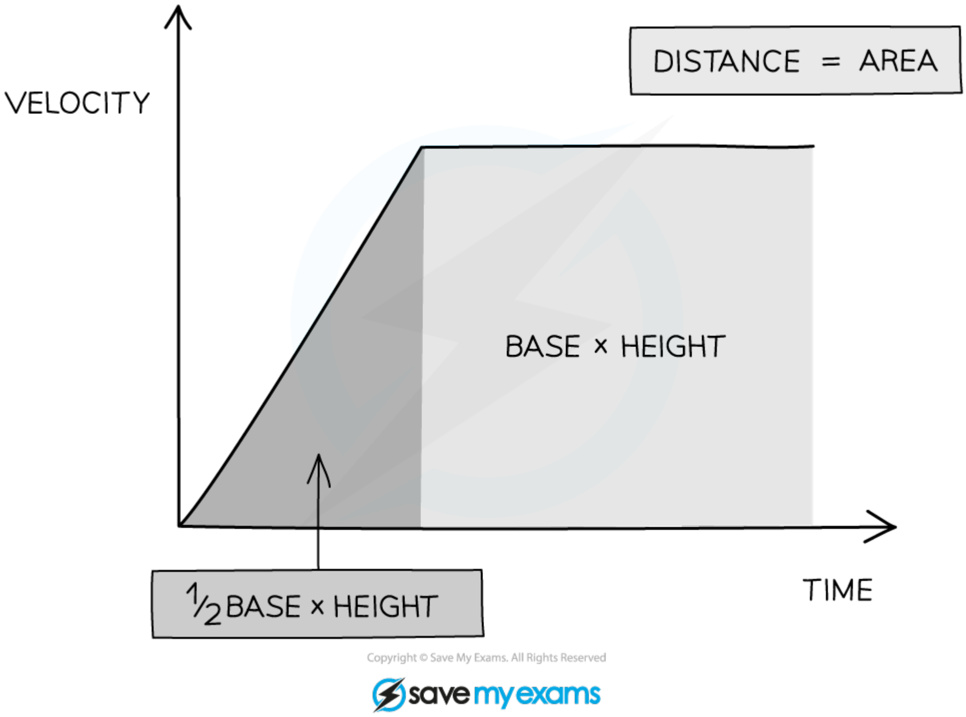
- If the line is **horizontal**, the **velocity is constant** (no acceleration)
- If the **line slopes upwards** then the object is **accelerating** (speeding up)
- If the **line goes down** then the object is **decelerating** (slowing down)

1. General Physics

YOUR NOTES

Calculating Distance

- The **distance travelled** by an object can be found by determining the **area beneath the graph**



The distance travelled can be found from the area beneath the graph

- If the area beneath the graph forms a triangle (the object is accelerating or decelerating) then the area can be determined using the formula:
- If the area beneath the graph is a rectangle (constant velocity) then the area can be determined using the formula:

area = ½ x base x height

- If the area beneath the graph is a rectangle (constant velocity) then the area can be determined using the formula:

area = base x height

1. General Physics

YOUR NOTES
⬇

Exam Tip

When asked to find the distance, start by stating:

`distance = area beneath graph`

A **common mistake** is to try and find distance using the distance-speed-time equation. This equation will not work if the speed of the object is changing.

1. General Physics

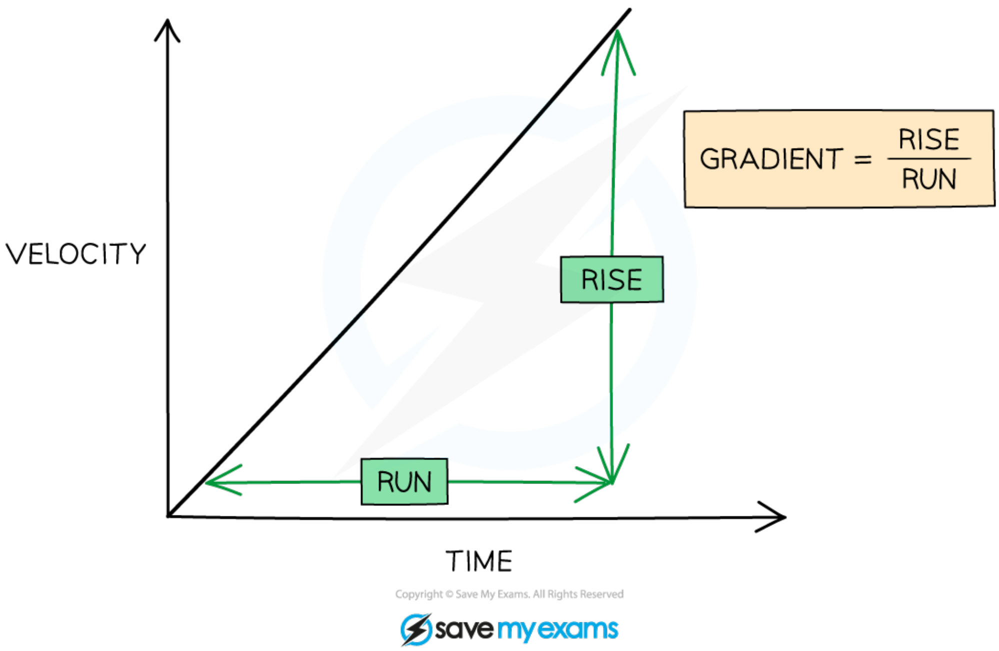
YOUR NOTES

Extended Only

Calculating Acceleration

- The **acceleration** of an object is given by the **gradient** of the graph:

$$acceleration = gradient = \frac{rise}{run}$$



Graph showing how acceleration can be determined from gradient

- Lines that slope downwards have **negative gradients** and so can be said to have **negative accelerations**: This is the **same thing as a deceleration**
- If the gradient of the line changes then the acceleration of the body must be changing:
  - A line with constant gradient represents constant acceleration (linear motion)
  - A curved line represents changing acceleration – either decreasing (if the gradient gets smaller) or increasing (if the gradient gets large)

1. General Physics

YOUR NOTES
⬇

Exam Tip

Remember to include units when giving your answers. The units of acceleration, for example, are m/s^2

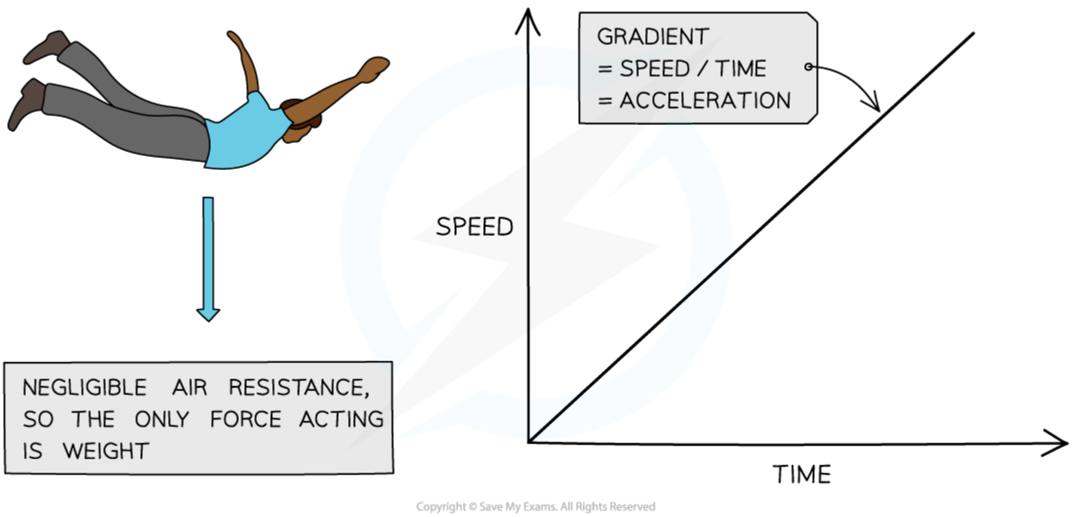
1. General Physics

YOUR NOTES

1.2.4 FREEFALL

Freefall: Basics

- In the absence of air resistance, all objects fall with the same acceleration, regardless of their mass
- This acceleration is equal to the gravitational field strength and is approximately 10 m/s^2 near the Earth's surface
- So long as air resistance remains insignificant, the speed of a falling object will increase at a steady rate, getting larger the longer it falls for.



In the absence of air resistance objects fall with constant acceleration

1. General Physics

YOUR NOTES ⬇

Extended Only

Terminal Velocity

- When a parachutist jumps out of an aeroplane, two main forces act:
    - **Weight (the force of gravity)**
    - **Air resistance**

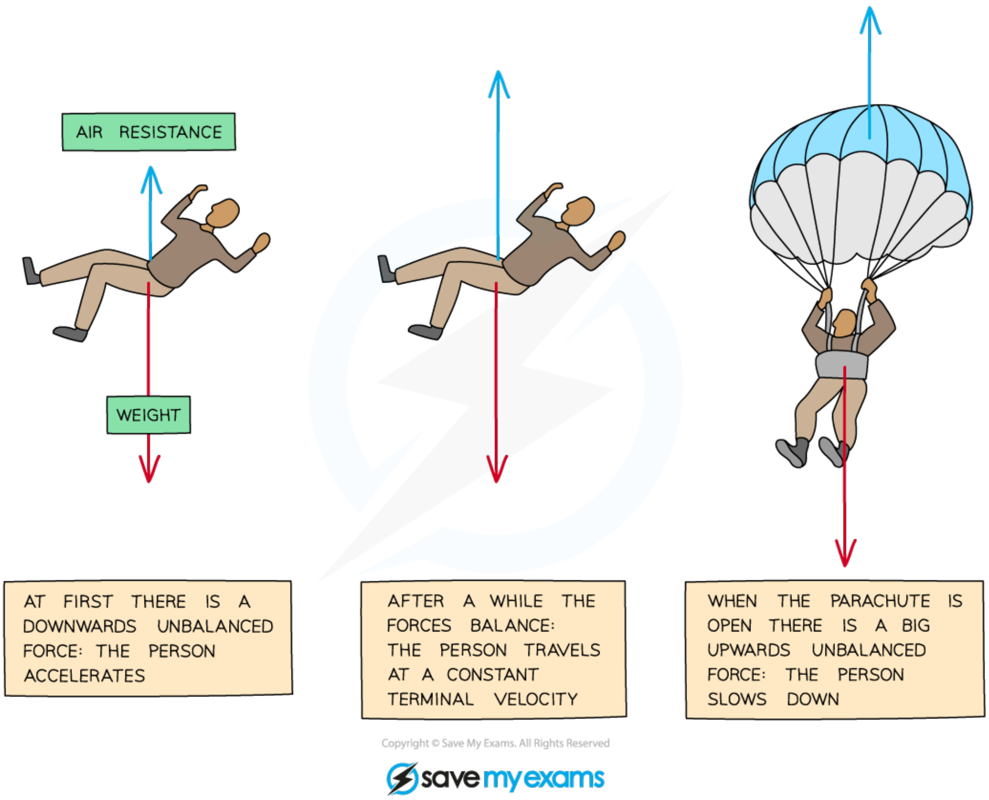
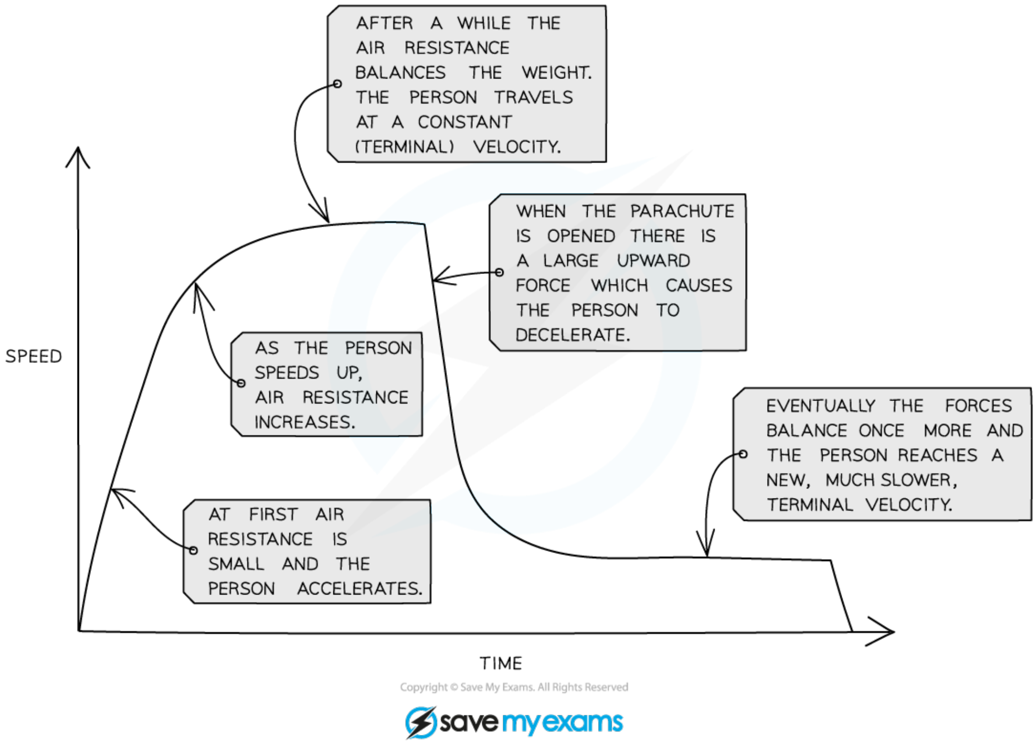


Diagram showing how the changing forces on a skydiver

1. General Physics

YOUR NOTES

- Initially the air resistance is very small. **There is a downwards unbalanced force** and the skydiver accelerates
- As the skydiver speeds up, **the air resistance increases**
- Eventually the air resistance balances the weight and so the skydiver travels at a constant speed – **terminal velocity**
- When the parachute is opened the increase air resistance on the parachute creates an upwards unbalanced force, making the parachuting the slow down



Graph showing how the velocity of a skydiver changes during the descent

1. General Physics

YOUR NOTES
⬇

Exam Tip

The force of gravity on an object is called weight. If asked to name this force make sure you use this word: Don’t refer to it as “gravity” as this term could also mean gravitational field strength and so would probably be marked wrong.

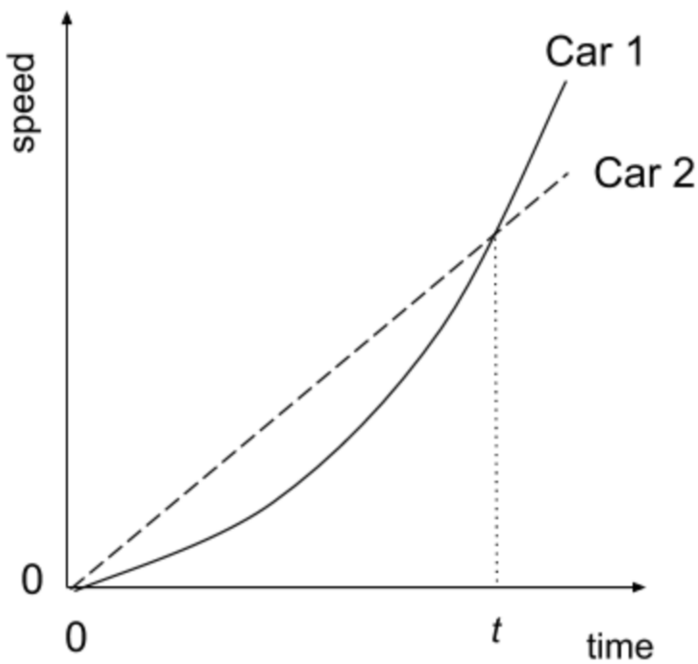
Likewise, refer to the upward force as air resistance or drag. The terms wind resistance and air pressure mean different things and so would also be marked wrong.

1. General Physics

YOUR NOTES

Exam Question: Easy

Two cars are in a race. The graph shows their speed as it changes with time.



What can be said about the motion of the two cars at time *t*?

A They have both travelled the same distance.

B Their acceleration is the same.

C Car 1 is overtaking car 2.

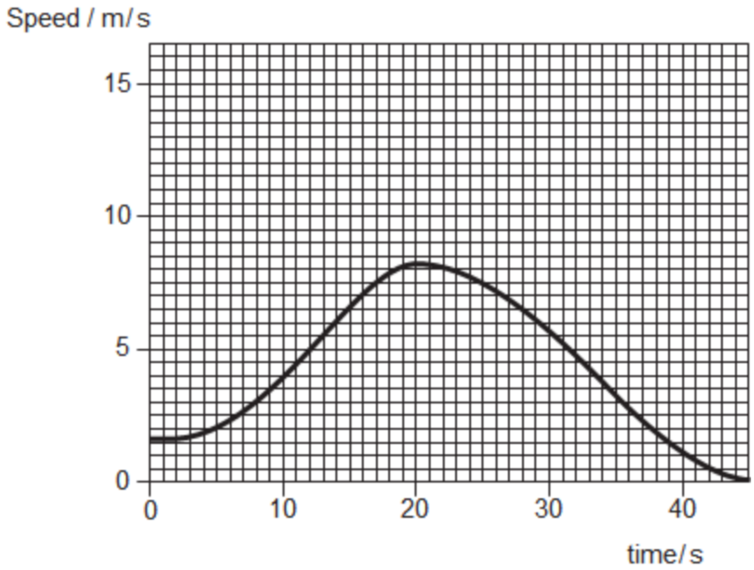
D Both cars are travelling at the same speed.

1. General Physics

YOUR NOTES

Exam Question: Medium

The graph shows the speed-time graph of a cyclist who is moving in a straight line.



What is the acceleration of the cyclist at a time of 20 seconds?

A 0.5 m/s^2

B -0.5 m/s^2

C 0 m/s^2

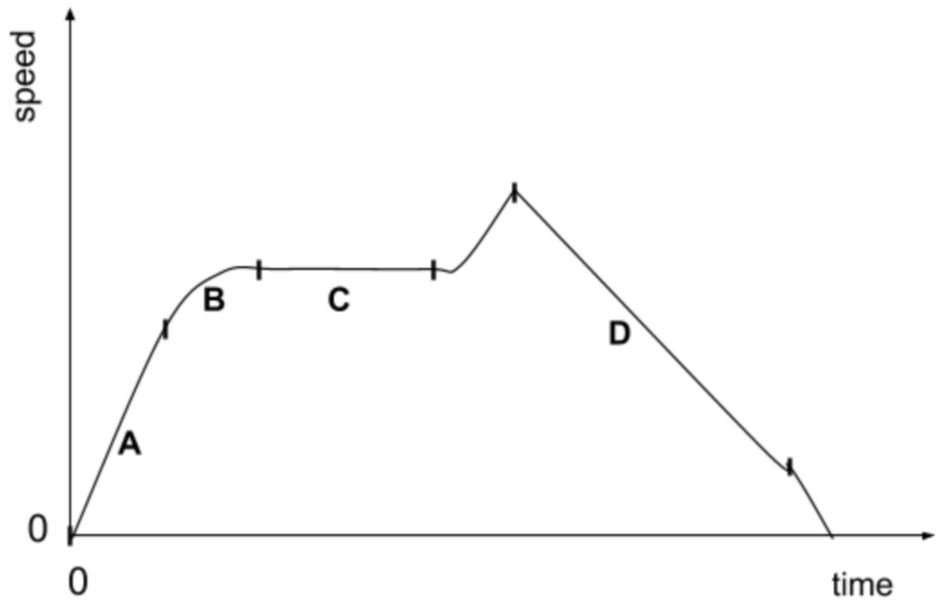
D 11.5 m/s^2

1. General Physics

YOUR NOTES ⬇

Exam Question: Hard

The graph below shows the journey of a truck along a straight, flat road.



During which part of the journey is the resultant force on the truck equal to zero?

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

YOUR NOTES
⬇

1.3 MASS & WEIGHT

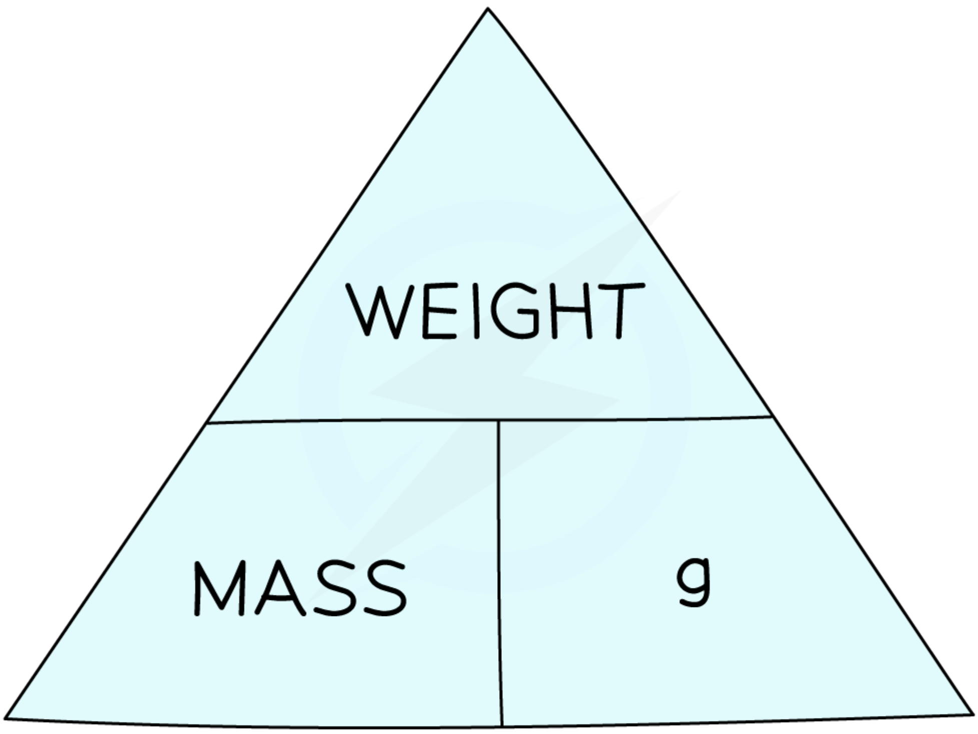
1.3.1 MASS & WEIGHT

Mass & Weight: Basics

- **Mass** (measured in kilograms, **kg**) is related to **the amount of matter in an object**
- **Weight** (measured in newtons, **N**) is **the force of gravity on a mass**
- The size of this force depends on the **gravitational field strength** (often called gravity, g, for short)

```
weight = mass x gravitational field strength
W = m x g
```

- You can rearrange this equation with the help of the formula triangle:



1. General Physics

YOUR NOTES

Use the formula triangle to help you rearrange the equation

- The value of g (the gravitational field strength) varies from planet to planet
- On Earth:

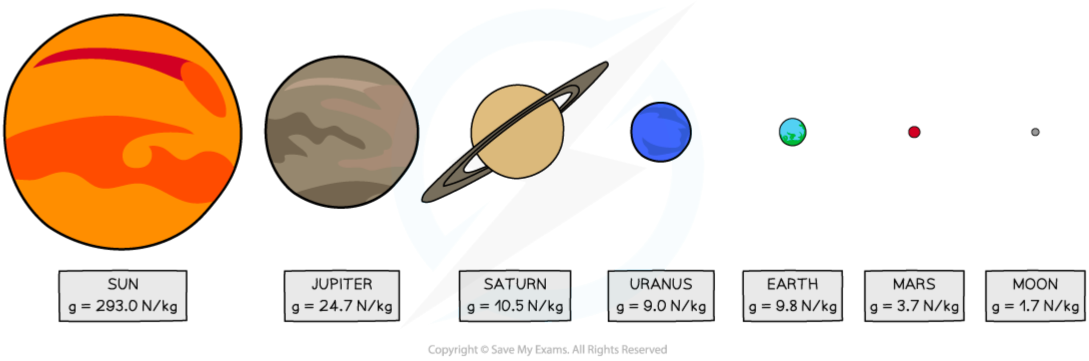
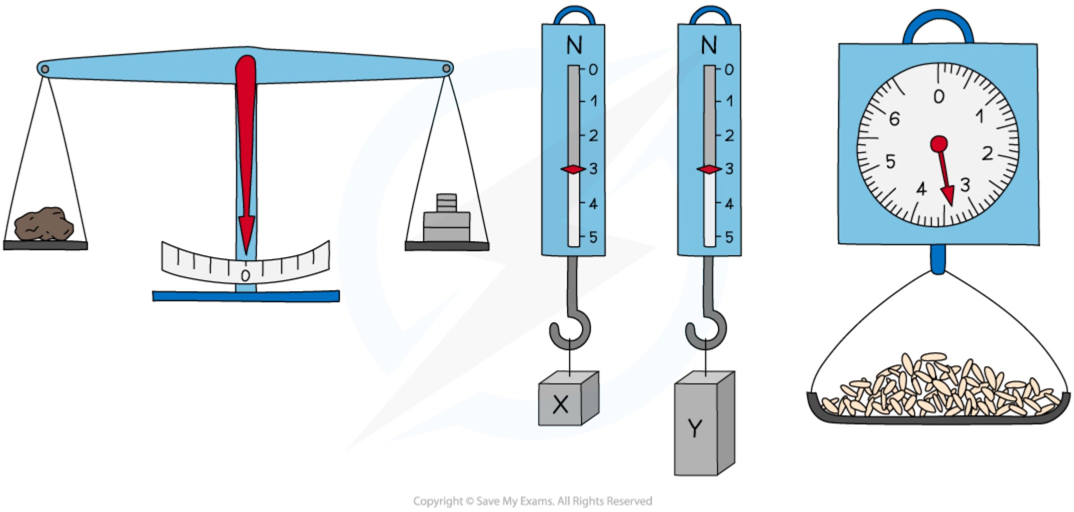


Diagram showing the gravitational field strengths of the planets in our solar system

- The weight (and hence mass) of two objects can be compared using a balance



A balance can be used to compare two different weights

1. General Physics

YOUR NOTES
⬇

Exam Tip

- Mass is usually measured in **kilograms** in Physics. If it is given in grams, you might need to convert to kilograms by dividing the mass by 1000
- It is easy to confuse mass and weight. – take care to use each term appropriately
- When referring to the force of gravity:
    - be careful to call it weight, gravitational force or gravitational attraction
    - Don't refer to it as just gravity and certainly don't call it gravitational field strength or gravitational potential (both of which mean different things)

1. General Physics

YOUR NOTES

Extended Only

The Significance of Mass

- Mass has two significant effects in Physics:
  - **The mass of an object's opposed any attempt to change that object's motion**
    The greater the mass of an object, the more difficult it is to speed it up, slow it down or change its direction
    This property of mass is sometimes referred to as **inertia**
  - **Mass is also the source of an object's weight** – the force of gravity on a mass
    The greater the mass, the greater the weight

Exam Question: Easy

Which of the quantities below is weight an example of?

A Field strength

B Acceleration

C Mass

D Force

1. General Physics

YOUR NOTES

Exam Question: Hard

A teacher has a hot cup of tea.

Some of the liquid in the cup evaporates.

Choose the line from the table below that correctly describes what happens to the mass and the weight of the tea in the cup.

	mass	weight
A	increases	decreases
B	increases	increases
C	decreases	decreases
D	decreases	increases

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

YOUR NOTES
⬇

1.4 DENSITY

1.4.1 DENSITY

Density: Basics

- **Density is the mass per unit volume** of a material:
  - Objects made from low-density materials typically have a low mass, whilst similar-sized objects made from high-density materials have a high mass
    (Think of how heavy a bag full of feathers is compared to a similar bag full of metal)
- Density is related to mass and volume by the following equation:

$$\text{DENSITY} = \frac{\text{MASS}}{\text{VOLUME}}$$

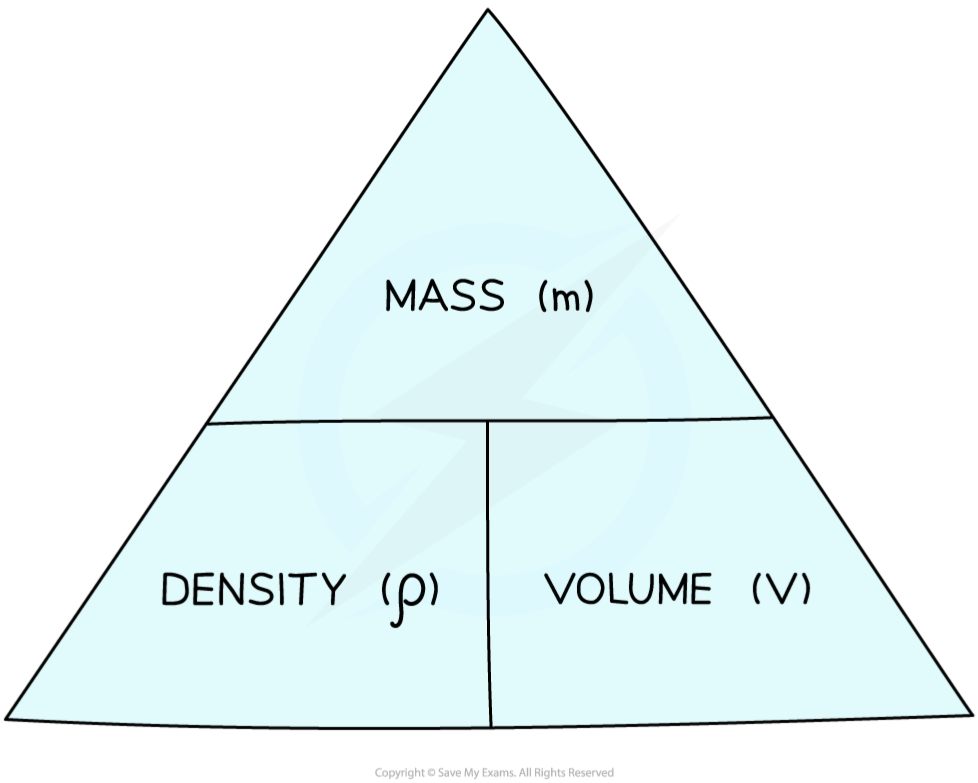
$$\rho = \frac{m}{V}$$

(Note: The greek letter ρ is used to mean density)

1. General Physics

YOUR NOTES

- You can rearrange this equation with the help of the formula triangle:



Use the formula triangle to help you rearrange the equation

- The units of density depend on what units are used for mass and volume:
  - If the mass is measured in ***g*** and volume in ***cm***^{3}, then the density will be in ***g/cm***^{3}
  - If the mass is measured in ***kg*** and volume in ***m***^{3}, then the density will be in ***kg/m***^{3}

1. General Physics

YOUR NOTES
⬇

Exam Tip

• The main thing to remember is that **density is mass per unit volume**

• In Physics, mass is almost always measured in kg

Density is the only topic in which physicists sometimes use grams instead

1. General Physics

YOUR NOTES

Floating

- In general, an object will float in a liquid if the average density of that object is less than the density of the liquid it is placed in
- Water, for example, has a density of about 1 g/cm^3
    - If an object has a density of less than 1 g/cm^3 then it will float in water
    - If an object has a density that is greater than 1 g/cm^3 then it will sink in water

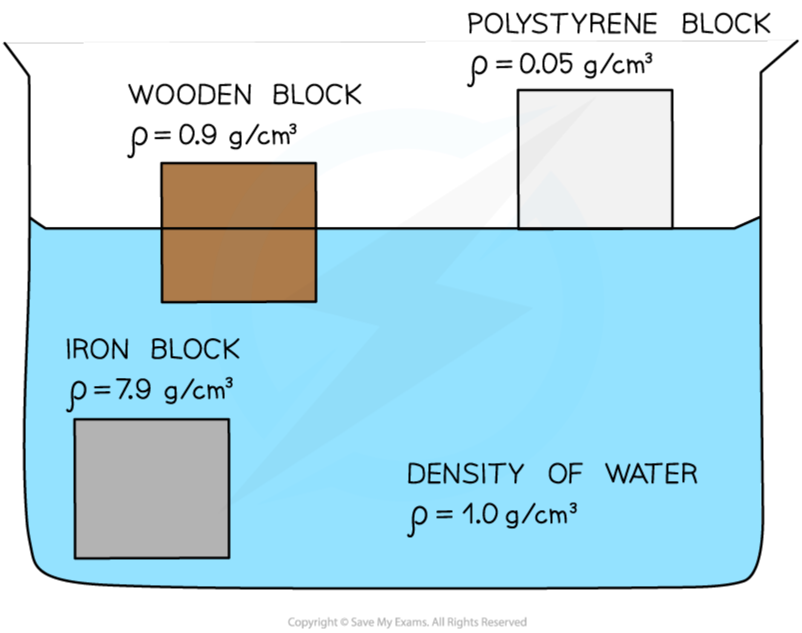


Diagram showing the relationship between an object's density and its ability to float in water

1. General Physics

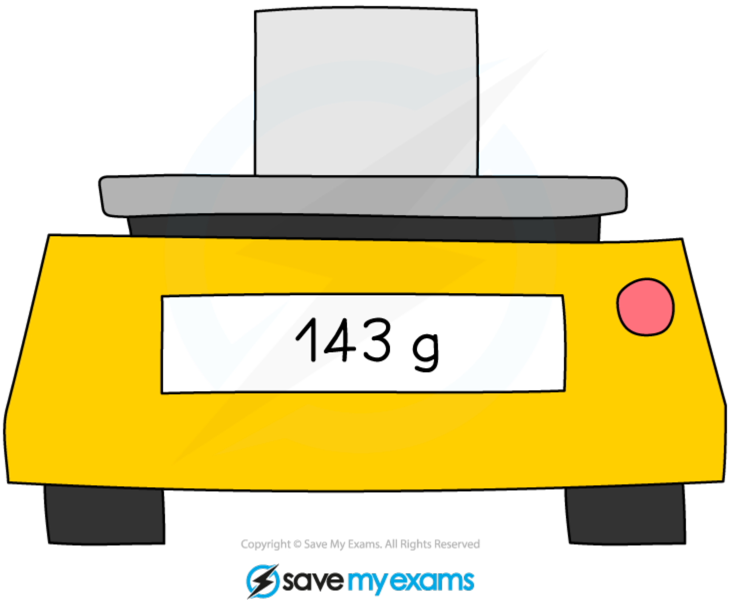
YOUR NOTES

Measuring Density

- To measure the density of an object, we must measure its mass and volume and then use the following equation:

$$\text{DENSITY} = \frac{\text{MASS}}{\text{VOLUME}}$$

- The mass of an object can be measured quite simply by placing it on a **top pan balance**
  You ought to state that you will 'zero' the balance before using it



Always zero a top pan balance before taking any measurements

- In the case of a liquid, the liquid must be placed in a container, the mass of which should be measured both when it is empty and when it contains the liquid:
  - The mass of the liquid will be the difference between the two values

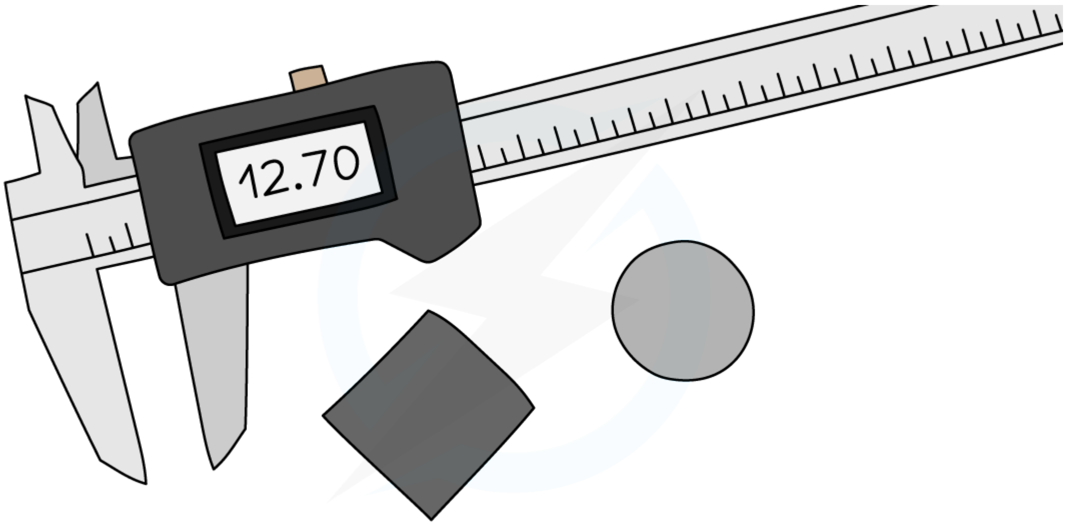
1. General Physics

YOUR NOTES

- The volume can be determined in a couple of ways:

Regular shapes (e.g. cubes, spheres, cylinders):

- The width (and length) can be measured using a ruler or a pair of **digital calipers**
- To make the measurements accurate, several measurements should be taken between different faces or points on the circumference, and an average taken



When measuring the width (or diameter) take several readings between different points and take an average

- The volume can then be calculated using an appropriate equation:

SPHERE: $V = \frac{4}{3}\pi r^3$

CUBE: $V = d^3$

CYLINDER: $V = \pi r^2 \times L$

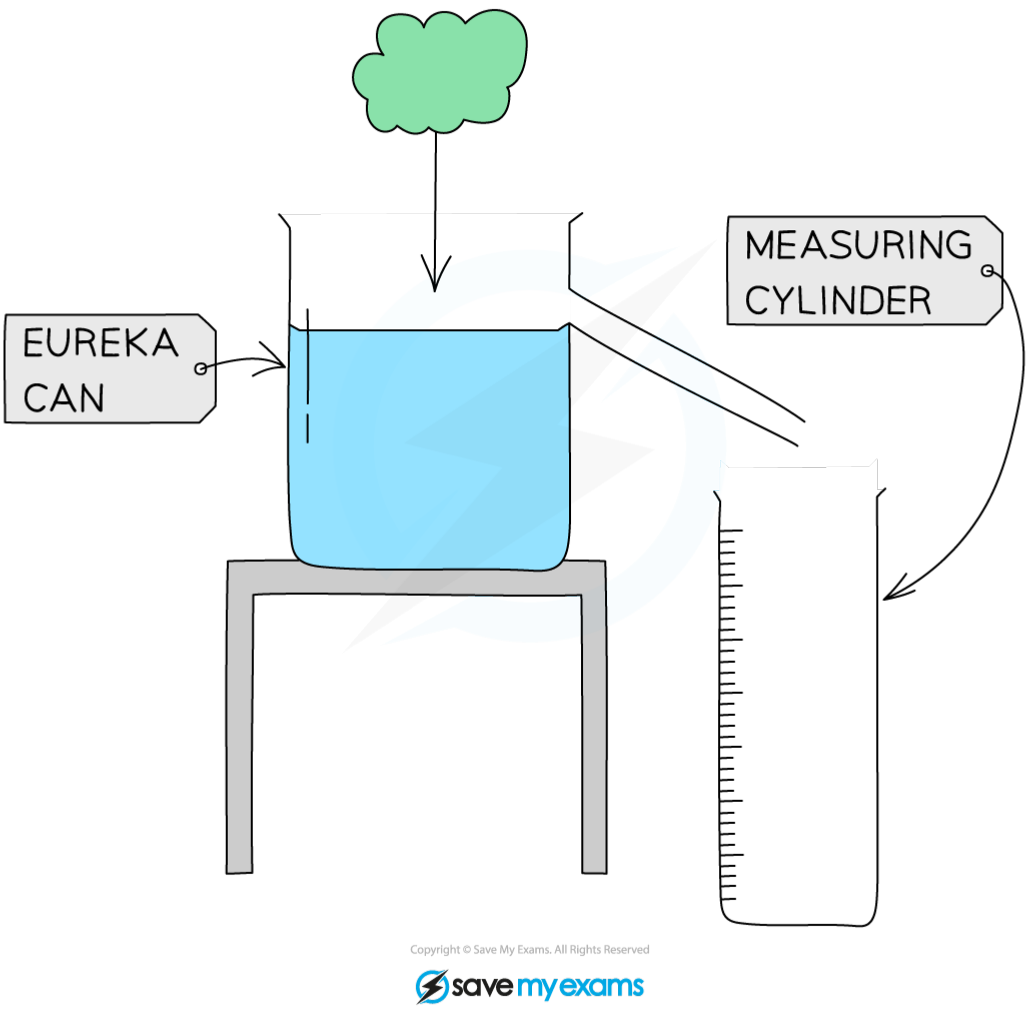
1. General Physics

YOUR NOTES

(Note: When measuring the width of a sphere or cylinder, divide the measurement by two to find the radius)

Irregular shapes:

- The volume can be found using a Eureka can:



Placing an object in a full Eureka can will displace water equal to its volume

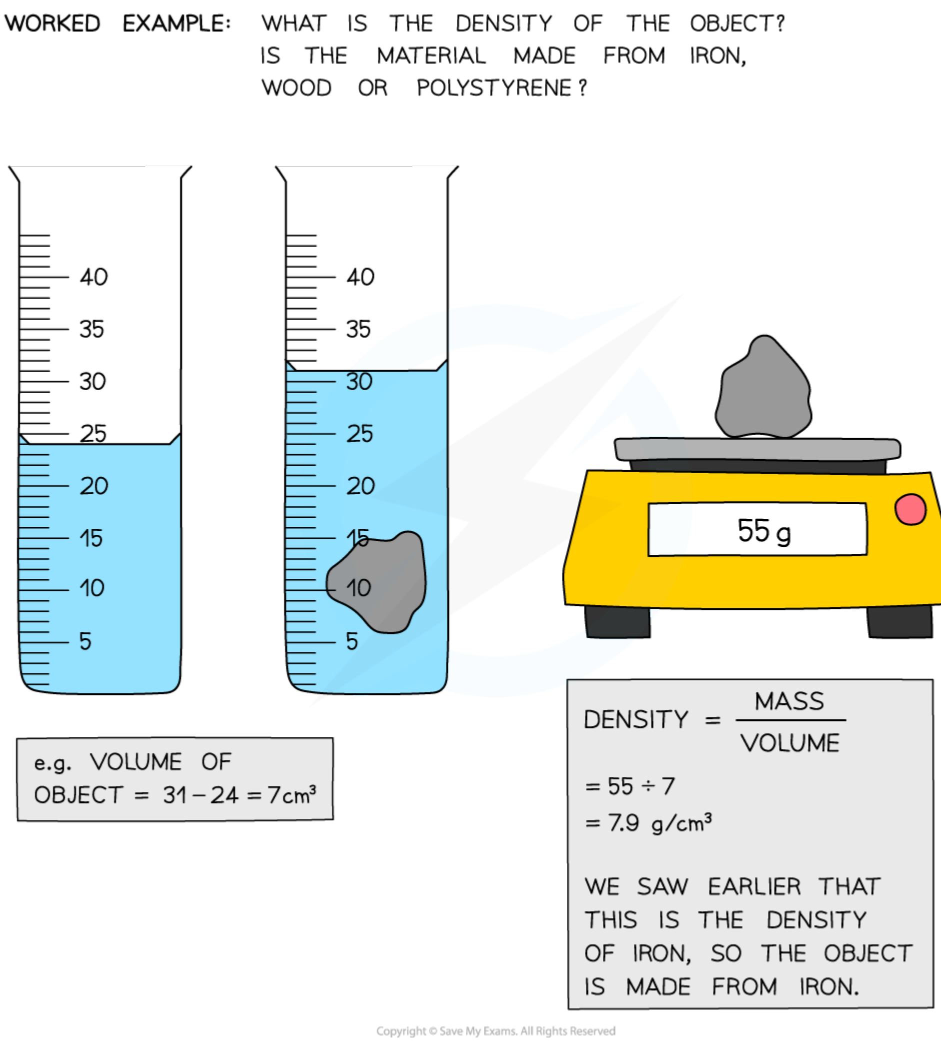
- Fill the Eureka can with water
- Place an empty measuring cylinder below its spout

1. General Physics

YOUR NOTES

- Now carefully lower the object into the Eureka can (use a piece of string, perhaps)
- Measure the volume of displaced water in the measuring cylinder

- Alternatively, the object can be placed in a measuring cylinder containing a known volume of liquid, and the change in volume then measured



When an irregular solid is placed in a measuring cylinder, the level of the liquid will rise by an amount equal to the volume of the solid

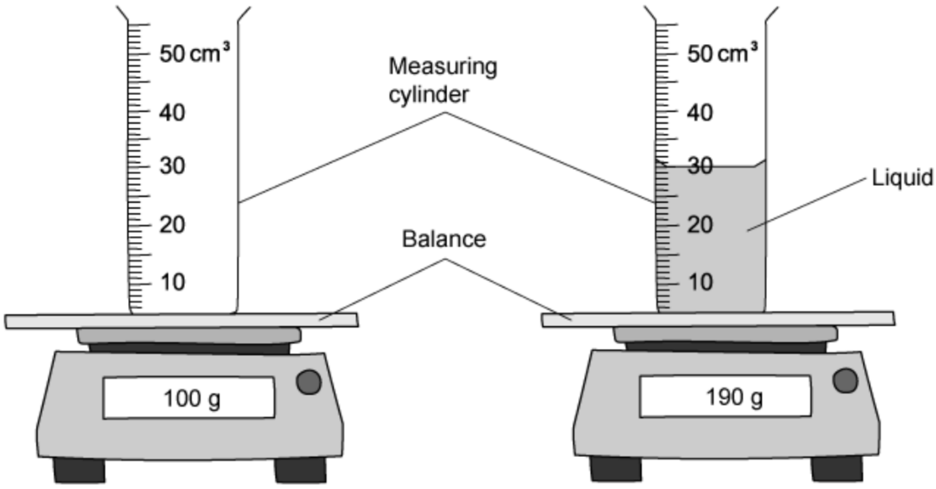
- Once the mass and volume of the shape is known, its density can be calculated

1. General Physics

YOUR NOTES

Exam Question: Hard

The diagram shows a measuring cylinder containing a liquid, and that same measuring cylinder when it is empty.



What is the density of the liquid?

A 6.30 g/cm^3

B 3.00 g/cm^3

C 0.33 g/cm^3

D 0.16 g/cm^3

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

YOUR NOTES
⬇

1.5 FORCES

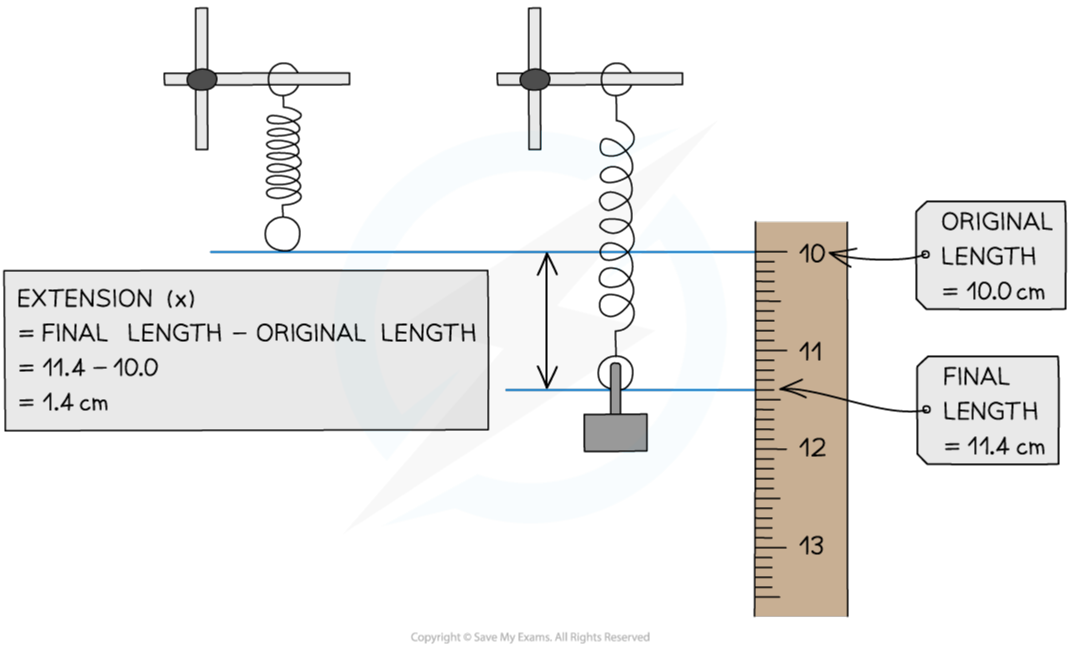
1.5.1 CHANGE OF SHAPE

1. General Physics

YOUR NOTES

Stretching Materials

- When forces are applied to materials, the size and shape of the material can change
- The method below describes a typical procedure for carrying out an investigation into the properties of a material



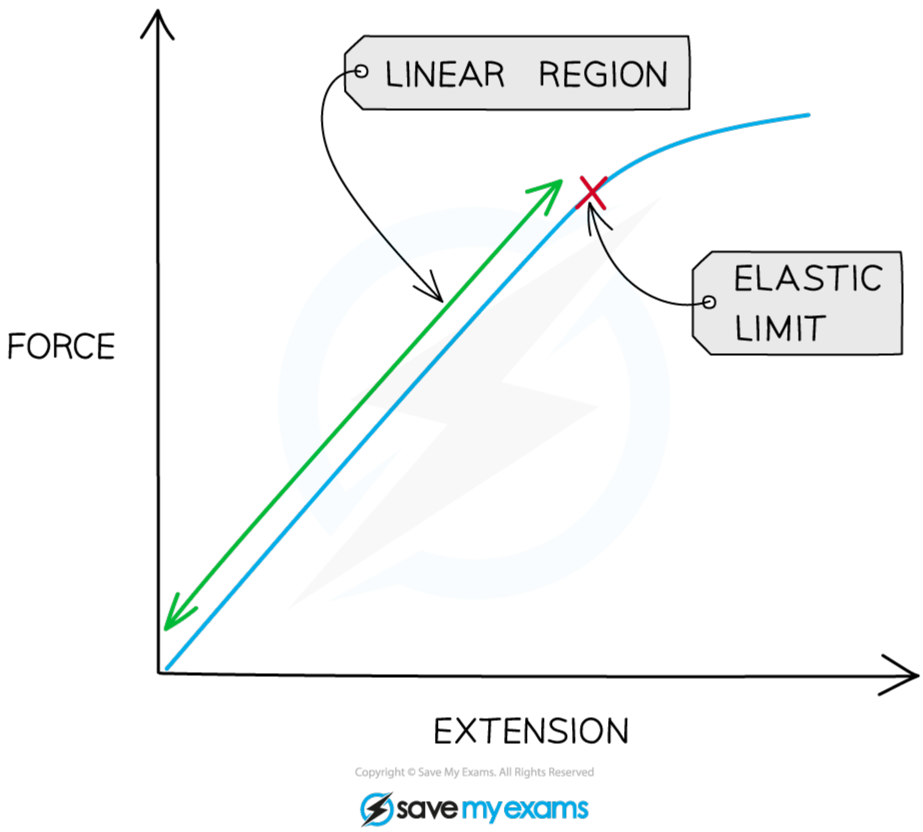
An experiment to measure the extension of a spring

- Set up the apparatus as shown in the diagram
- A single mass (0.1 kg, 100g) is attached to the spring, with a pointer attached to the bottom, and the position of the spring is measured against the ruler
- The mass (in kg) and position (in cm) are recorded in a table
- A further mass is added and the new position measured
- The above process continues until a total of 7 masses have been added
- The masses are then removed and the entire process repeated again, until it has been carried out a total of three times, and averages can then be taken

1. General Physics

YOUR NOTES ⬇

- Once measurements have been taken:
  - The force on the spring can be found by multiplying the mass on the spring (in kg) by 10 N/kg (the gravitational field strength)
  - The extension of the spring can be found by subtracting the original position of the spring from each of the subsequent positions
- Finally, a graph of extension (on the y-axis) against force (on the x-axis) should be plotted



A graph of force against extension for a metal spring

1. General Physics

YOUR NOTES
⬇

Extended Only

Hooke's Law

- Hooke's law states that:
  - **The extension of a spring is proportional to the applied force**

$$force = spring\ constant \times extension$$

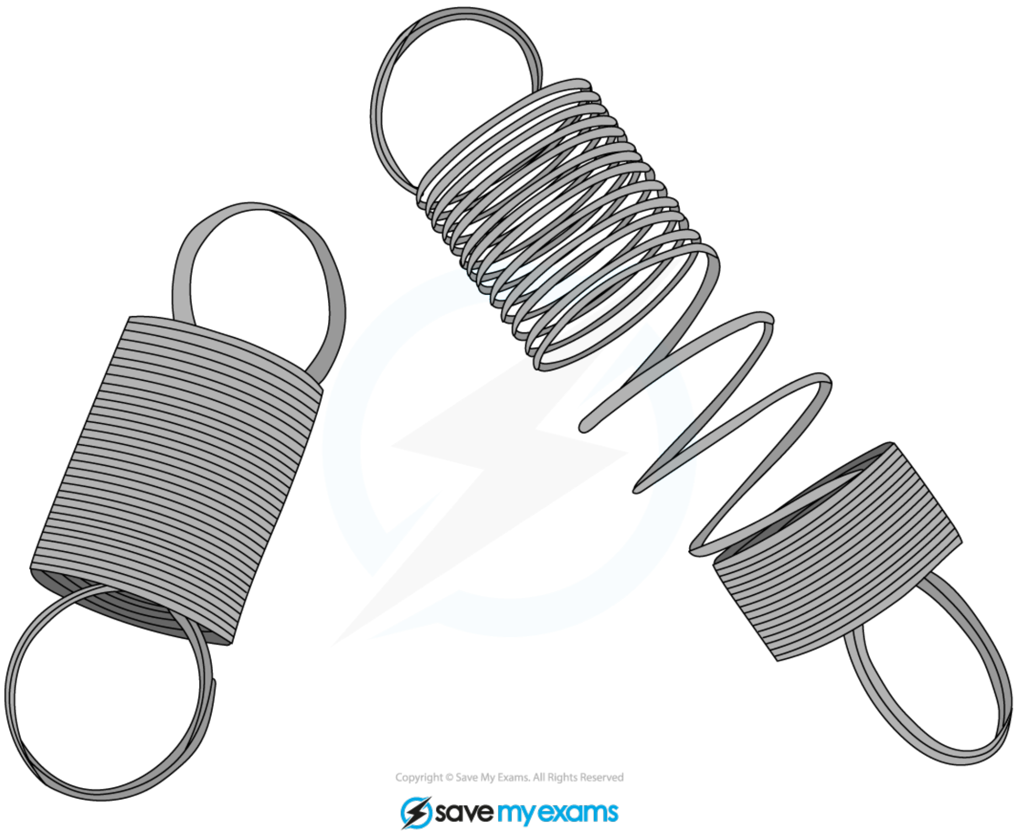
$$F = kx$$

(where k is the spring constant, which represents how stiff a spring is)

- Many other materials (such as metal wires) also obey Hooke's law
- Hooke's law is associated with the initial linear (straight) part of a force-extension graph
- Objects that obey Hooke's law will return to their original length after being stretched
- If an object continues to be stretched it can be taken past the **limit of proportionality** (sometimes called the elastic limit). At this point the object will no longer obey Hooke's law and will not return to its original length

1. General Physics

YOUR NOTES



The spring on the right has been stretched beyond the limit of proportionality

Exam Tip

A relationship is said to be **proportional** if the graph is a **straight line going through the origin.**

If a graph is a straight line but **does not go through the origin** the relationship is said to be **linear.**

1. General Physics

YOUR NOTES
⬇

1.5.2 BALANCED FORCES

Resultant Force

- When several forces act on a body, the resultant (overall) force on the body can be found by adding together forces which act in the same direction and subtracting forces which act in opposite directions:

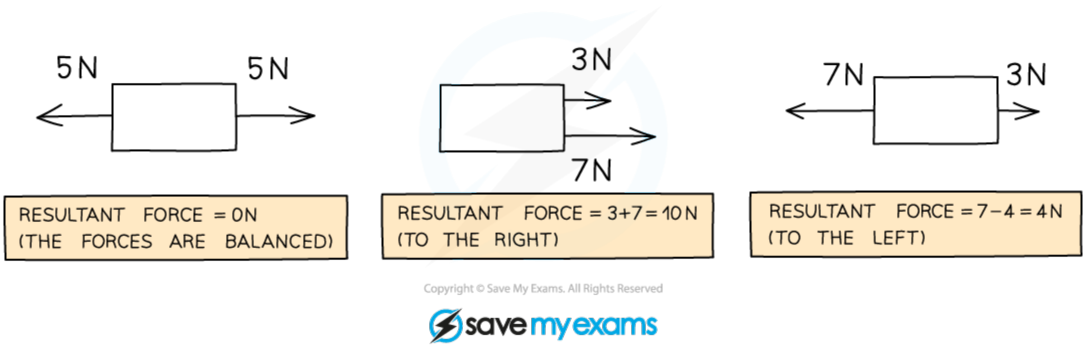
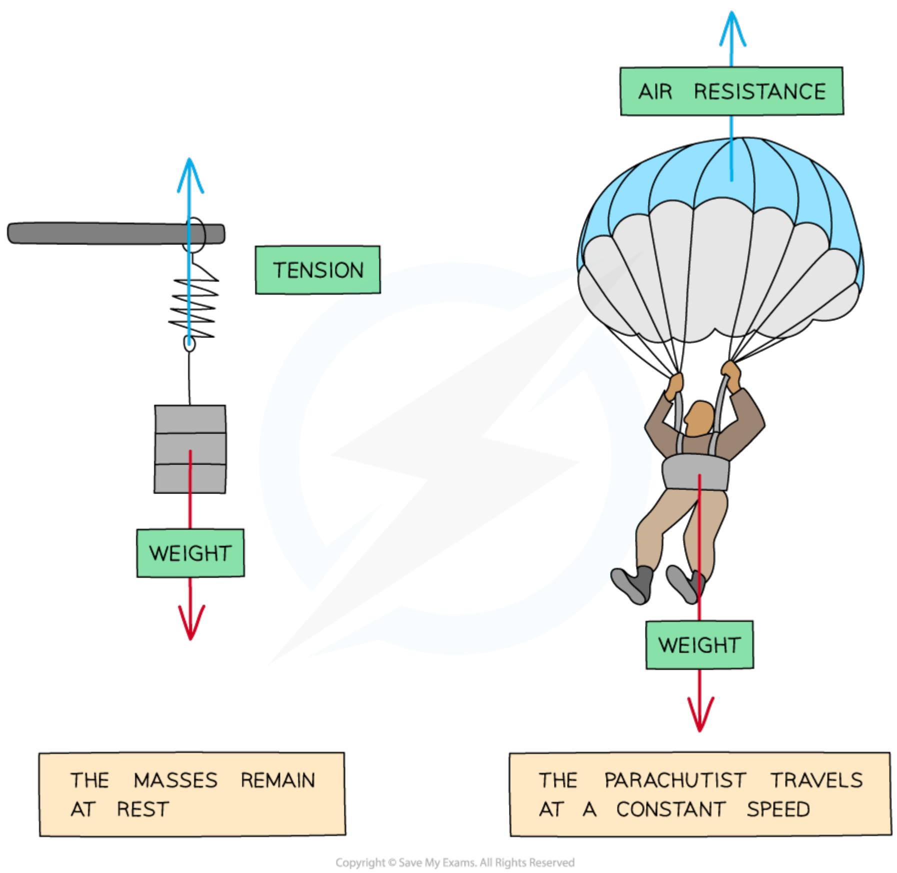


Diagram showing the resultant forces on three different objects

- When the forces acting on a body are balanced (i.e. there is no resultant force), the body will either remain at rest or continue to move in a straight line at a constant speed

1. General Physics

YOUR NOTES
⬇



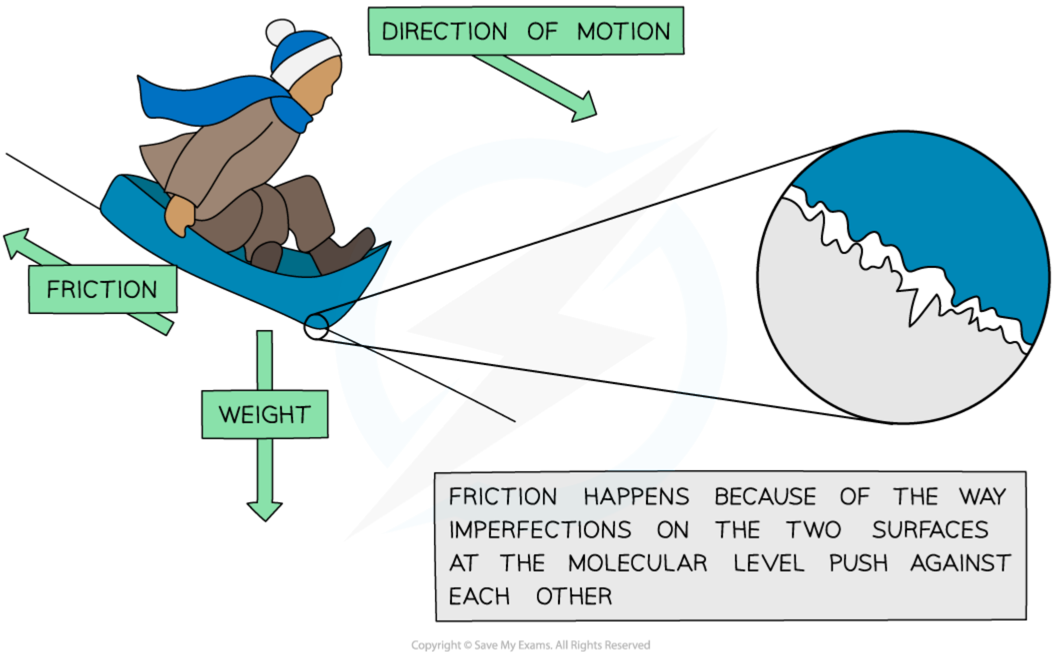
When the forces acting on a body are balanced the body will remain at rest or continue to travel at a constant speed in a straight line

1. General Physics

YOUR NOTES

Friction

- **Friction is a force that opposes the motion of an object** caused by the contact (rubbing) of two surfaces. It always acts in the opposite direction to the direction in which the object is moving



Friction opposes the motion of an object

- **Air resistance** (sometimes called drag) is a form of friction caused by a body moving through the air
- Friction (including air resistance) results in energy loss due to the transfer of energy from **kinetic to internal** (heat)

1. General Physics

YOUR NOTES
⬇

Exam Tip

The **resultant force** is sometimes also known as the **net force** or the **unbalanced force**.

Avoid referring to air resistance as *wind resistance* or *air pressure* – these are incorrect terms and will lose you marks if you use them when you actually mean air resistance.

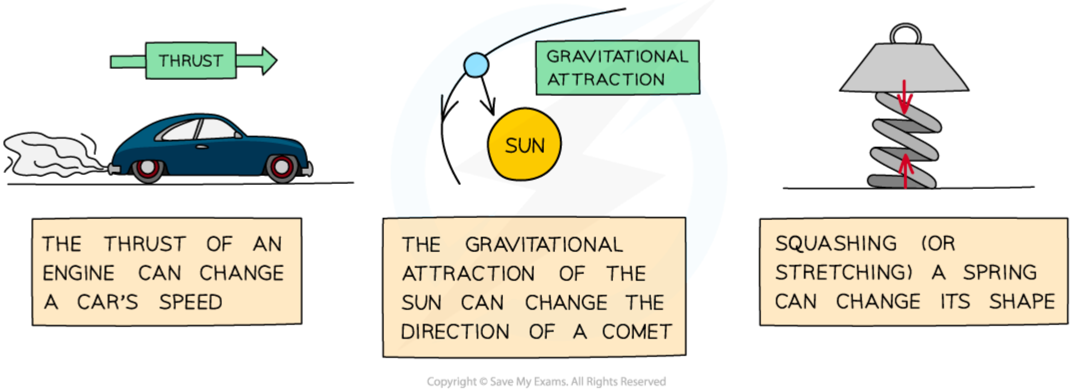
1. General Physics

YOUR NOTES

1.5.3 UNBALANCED FORCES

Forces & Motion

- When an **unbalanced (resultant) force** acts on an object, it can affect its motion in a number of ways:
  - The object could speed up
  - The object could slow down
  - The object could change direction



A resultant force can cause an object to speed up, slow down or change direction

1. General Physics

YOUR NOTES

Extended Only

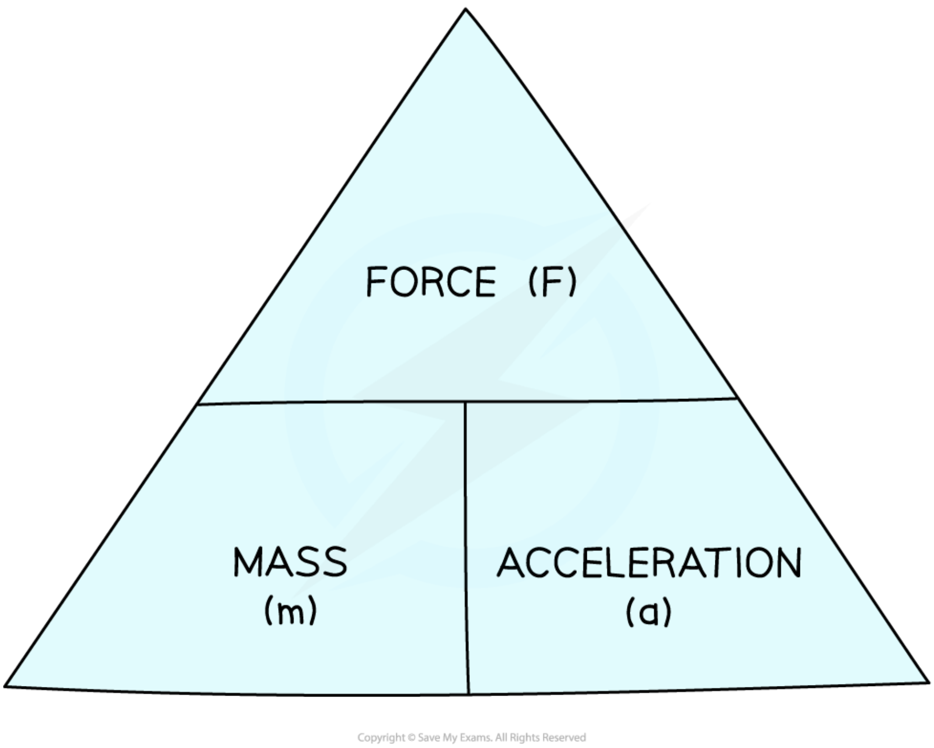
Acceleration

- Force, mass and acceleration are related by the following equation:

force = mass × acceleration

f = m × a

- You can rearrange this equation with the help of the formula triangle:



Use the formula triangle to help you rearrange the equation

- **The *greater* the force, the *greater* the acceleration** (for a given mass)
- For a given force, **the *smaller* the mass the *greater* the acceleration**

1. General Physics

YOUR NOTES

Exam Tip

If you are trying to find the acceleration check that you know both the unbalanced (resultant) force and the mass of the object. If you don't, you might need to calculate the acceleration using a different equation.

1. General Physics

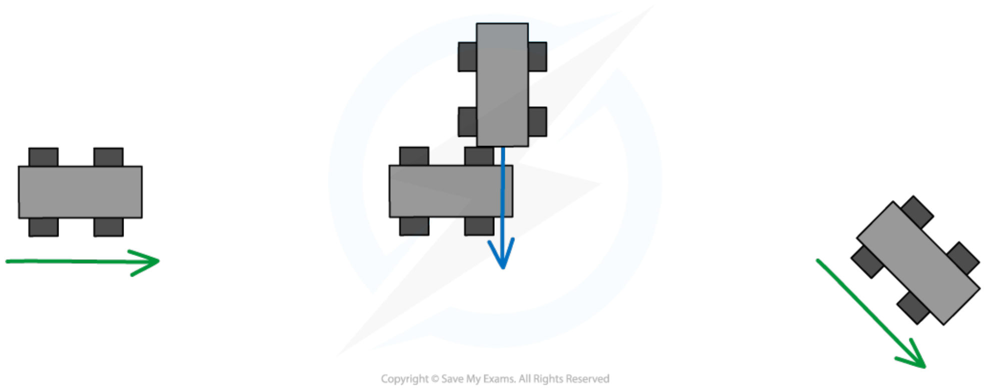
YOUR NOTES

1.5.4 CIRCULAR MOTION

Extended Only

Changing Direction

- **When a force acts at 90 degrees** to an object's direction of travel, the force will cause that object to change direction

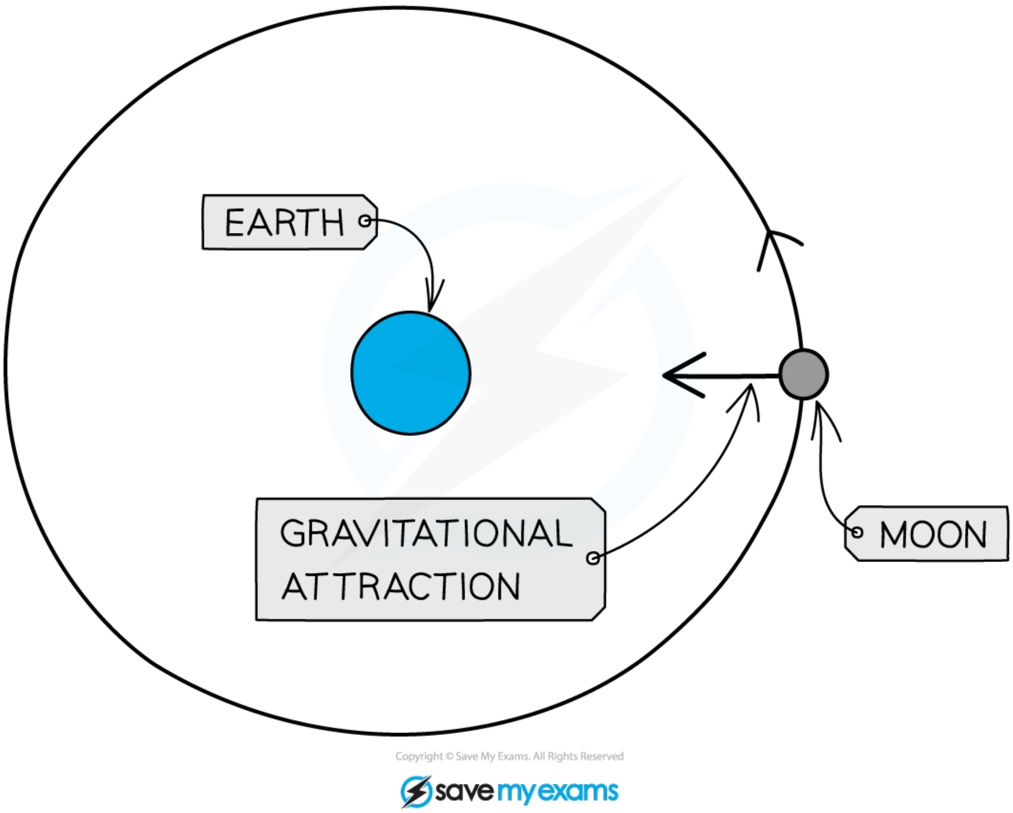


When the two cars collide, the first car changes its direction in the direction of the force

1. General Physics

YOUR NOTES

- If the force continues to act at 90 degrees to the motion, the object will keep changing its direction (whilst remaining at a constant speed) and travel in a circle
- This is what happens when a planet orbits a star (or satellite orbits a planet)



The Moon is pulled towards the Earth (at 90 degrees to its direction of travel). This causes it to travel in a circular path

- The force needed to make something follow a circular path depends on a number of factors:
  - **The mass of the object** (a greater mass requires a greater force)
  - **The speed of the object** (a faster-moving object requires a greater force)
  - **The radius of the circle** (a smaller radius requires a greater force)

1. General Physics

YOUR NOTES

1.5.5 TURNING EFFECT

The Moment of a Force

- **A moment is the turning effect of a force**
- Moments occur when forces cause objects to rotate about some pivot
- The size of the moment depends upon:
  - The size of the force
  - The distance between the force and the pivot
- The moment of a force is given by the equation:

```
Moment = Force × perpendicular distance from the pivot
```

- **Moments have the units** newton centimetres (**N cm**) or newton metres (**N m**), depending on whether the distance is measured in metres or centimetres

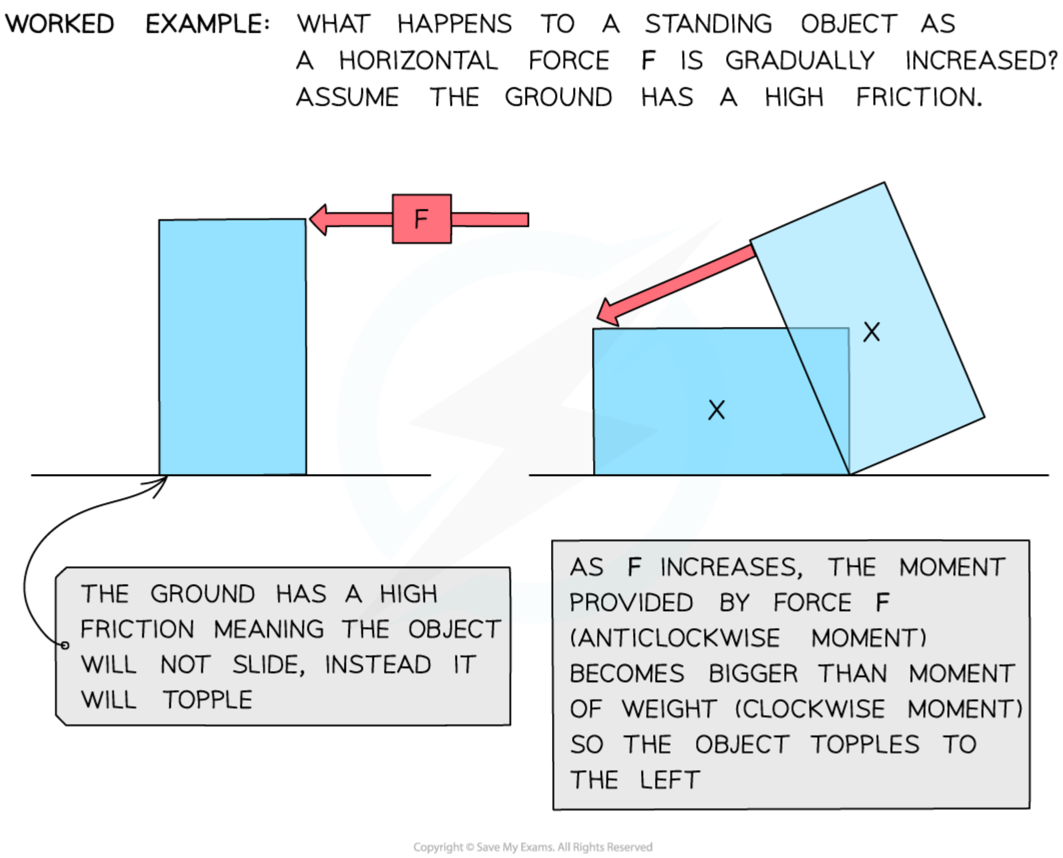


Diagram showing the moment of a force causing a block to topple

1. General Physics

YOUR NOTES
⬇

- Some other examples involving moments include:
    - Using a crowbar to prize open something
    - Turning a tap on or off
    - Opening or closing a door

1. General Physics

YOUR NOTES ⬇

The Principle of Moments

- The principle of moments states that:
  - **For a system to be balanced, the sum of clockwise moments must be equal to the sum of anticlockwise moments**

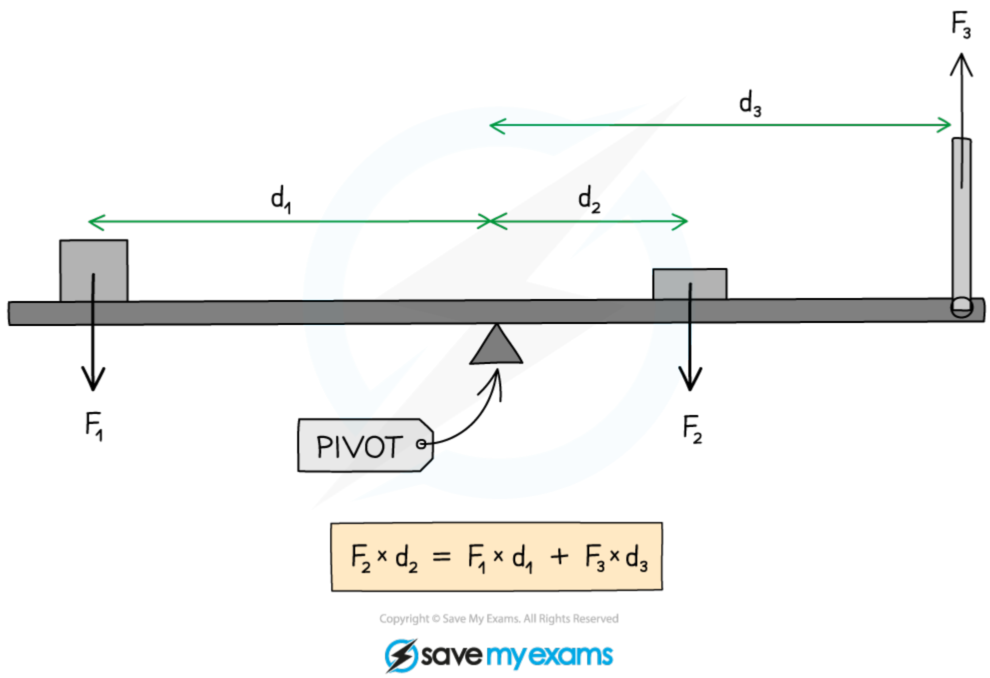


Diagram showing the moments acting on a balanced beam

- In the above diagram:
  - Force F_2 is supplying a clockwise moment;
  - Forces F_1 and F_3 are supplying anticlockwise moments
- Hence:

$$F_2 \times d_2 = F_1 \times d_1 + F_3 \times d_3$$

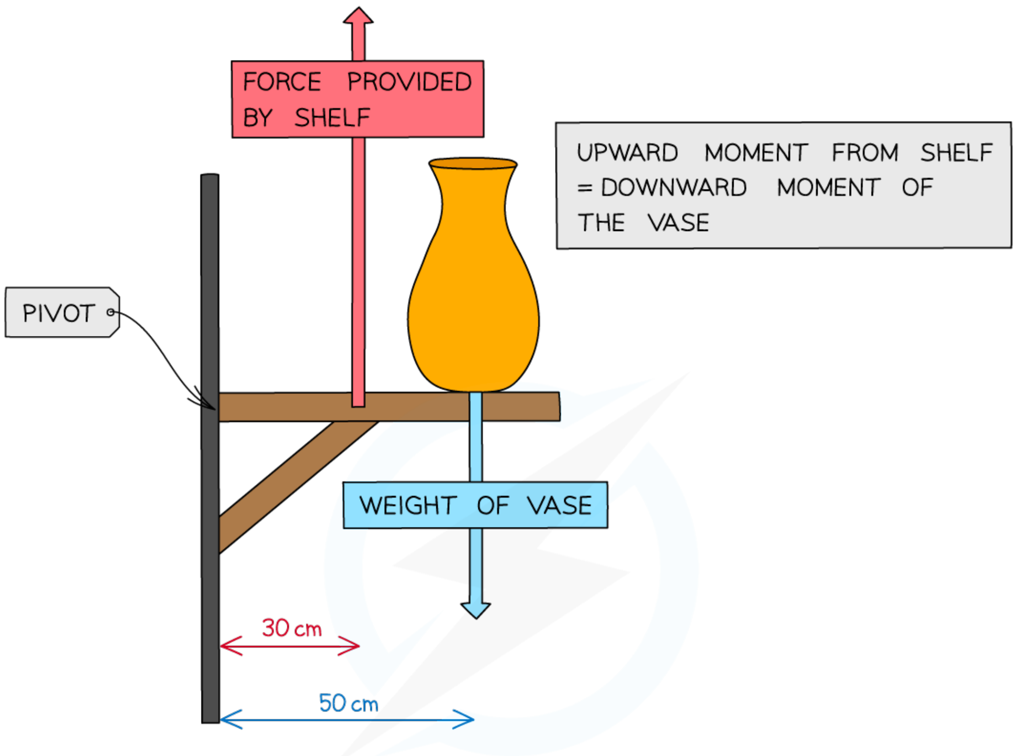
1. General Physics

YOUR NOTES
⬇

Extended Only

Example of The Principle of Moments

- The principle of moments doesn't just apply to seesaws – it is important in many other situations as well such as, for example, a shelf:



WORKED EXAMPLE: IF THE WEIGHT OF THE VASE IS 80N WHAT IS THE FORCE PROVIDED BY THE SHELF ?

$F_1 \times d_1 = F_2 \times d_2$

$F \times 30\,cm = W \times 50\,cm$

$F \times 0.3\,m = 40 \times 0.5\,m$

$F = 130\,N$

REMEMBER TO CONVERT cm TO m!

To prevent the shelf from collapsing, the support must provide an upward moment equal to the downward moment of the vase

1. General Physics

YOUR NOTES
⬇

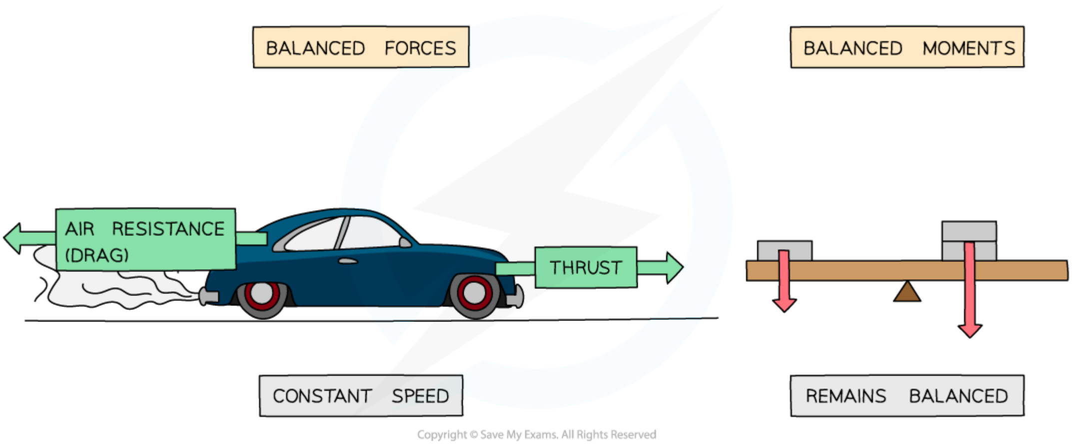
1. General Physics

YOUR NOTES
⬇

1.5.6 CONDITIONS FOR EQUILIBRIUM

Equilibrium Defined

- The term “equilibrium” means that an object keeps doing what it’s doing, without any change
- Therefore:
  - If the object is moving it will continue to move (in a straight line)
  - If it is stationary it will remain stationary
  - The object will also not start or stop turning
- The above conditions require two things:
  - **The forces on the object must be balanced** (there must be no resultant force)
  - **The sum of clockwise moments on the object must equal the sum of anticlockwise moments** (the principle of moments)



When the forces and moments on an object are balanced, the object will remain in equilibrium

- If the above two conditions are met, then **the object will be in equilibrium**

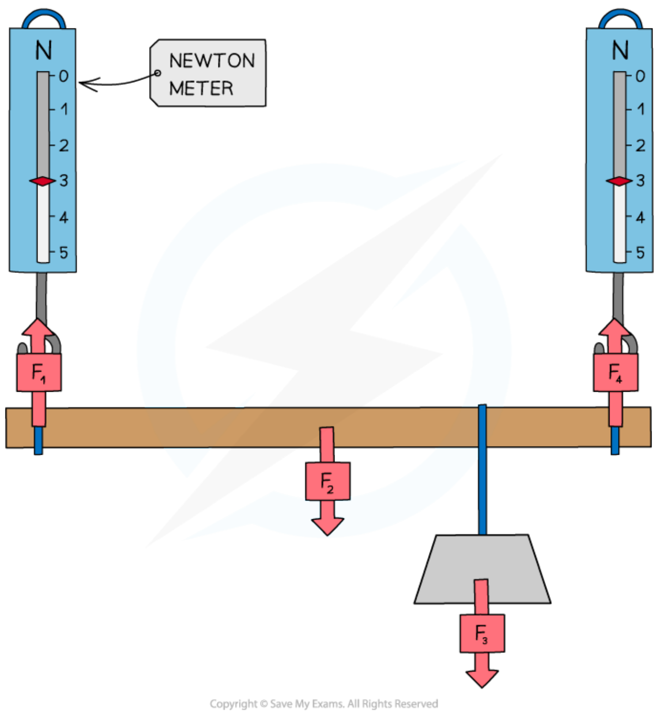
1. General Physics

YOUR NOTES

Extended Only

Demonstrating Equilibrium

- A simple experiment to demonstrate that there is no net moment on an object in equilibrium involves taking an object, such as a beam, and replacing the supports with newton (force) meters:



Several forces act on a supported beam, including the mass of the beam and the mass of an object suspended from it

- The beam in the above diagram is in equilibrium
- The various forces acting on the beam can be found either by taking readings from the newton meters or by measuring the masses (and hence calculating the weights) of the beam and the mass suspended from the beam
- The distance of each force from the end of the ruler can then be measured, allowing the moment of each force about the end of the ruler to be calculated
- It can then be shown that the sum of clockwise moments (due to forces F_2 and F_3) equal the sum of anticlockwise moments (due to forces F_1 and F_4)

1. General Physics

YOUR NOTES

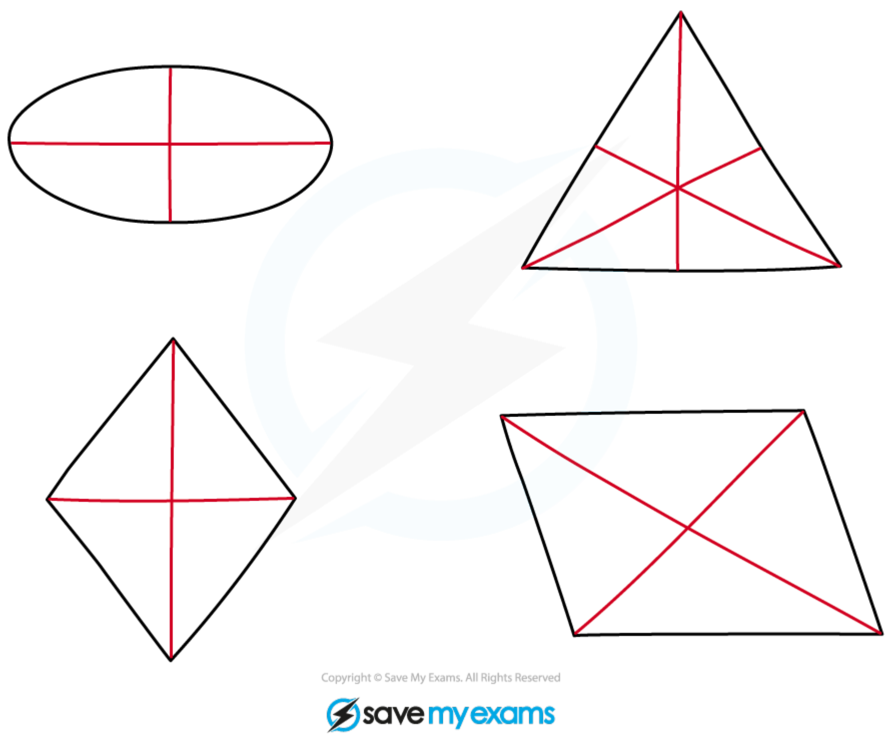
1. General Physics

YOUR NOTES

1.5.7 CENTRE OF MASS

Finding the Centre of Mass

- The centre of mass of an object (sometimes called the centre of gravity) is **the point through which the weight of that object acts**
- For a symmetrical object of uniform density (such as a symmetrical cardboard shape) **the centre of mass is located at the point of symmetry**:



The centre of mass of a regular shape can be found by symmetry

1. General Physics

YOUR NOTES

- **When an object is suspended from a point, the object will always settle so that it's centre of mass comes to rest below the pivoting point**
- This can be used to find the centre of mass of an irregular shape:

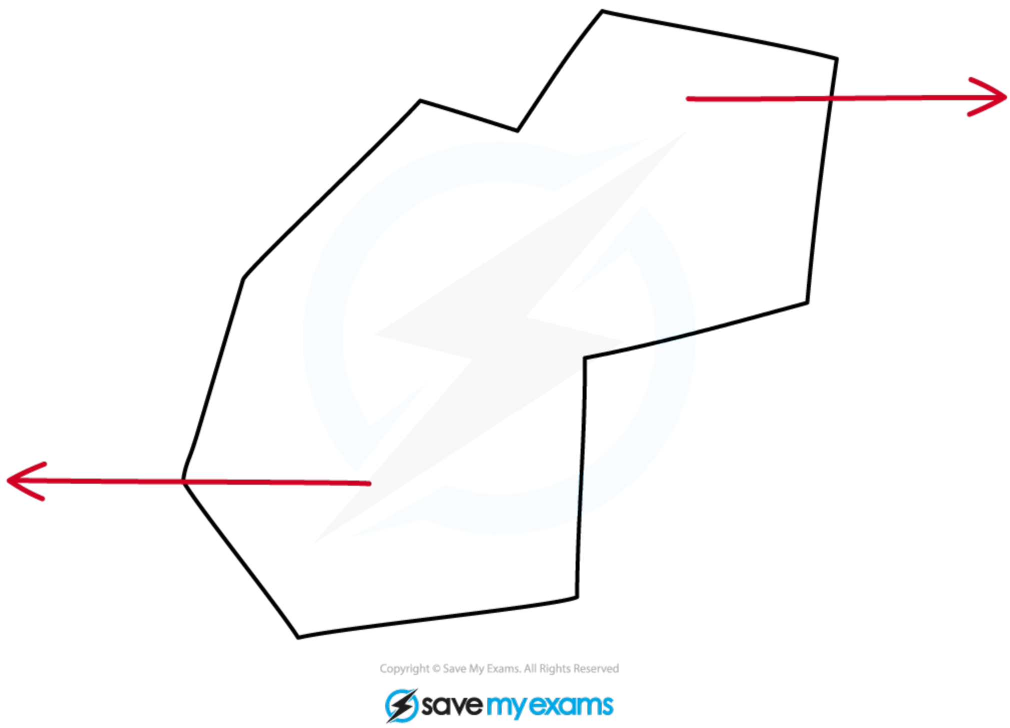


Diagram showing an experiment to find the centre of mass of an irregular shape

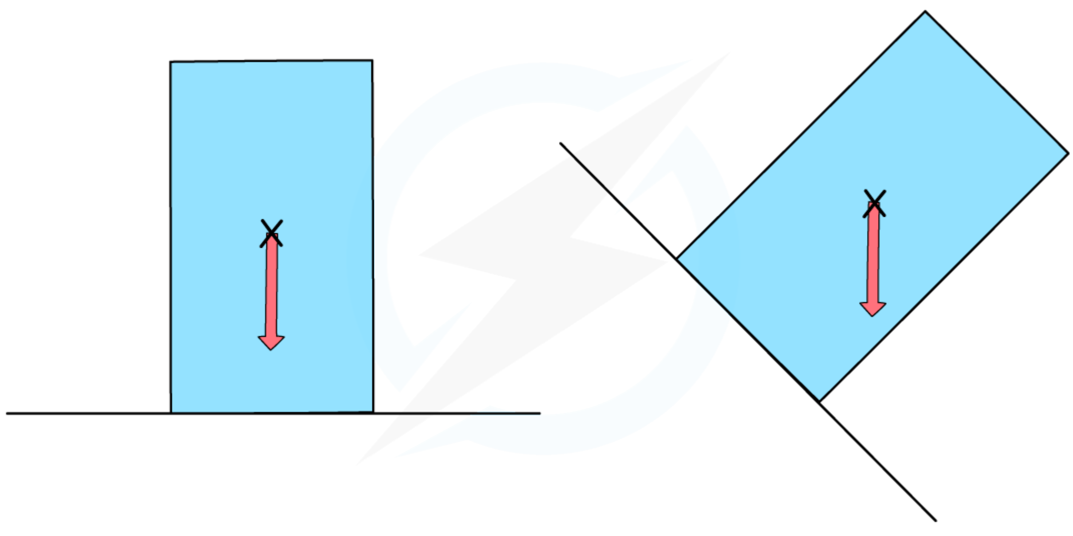
- The irregular shape (a plane laminar) is suspended from a pivot and allowed to settle
- A plumb line (lead weight) is then held next to the pivot and a pencil is used to draw a vertical line from the pivot (the centre of mass must be somewhere on this line)
- The process is then repeated, suspending the shape from two different points
- The centre of mass is located at the point where all three lines cross

1. General Physics

YOUR NOTES

Stability

- An object is stable when its centre of mass lies above its base

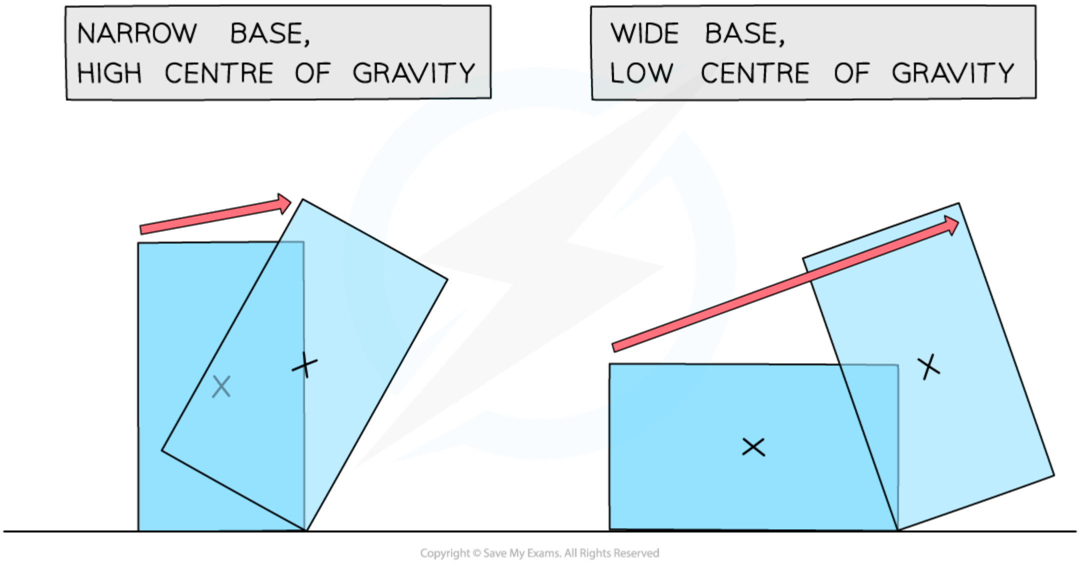


The object on the right will topple, as its centre of mass is no longer over its base

1. General Physics

YOUR NOTES

- If the centre of mass does not lie above its base, then an object will topple over
- The most stable objects have a low centre of mass and a wide base



The most stable objects have wide bases and low centres of mass

1. General Physics

YOUR NOTES

1.5.8 SCALARS & VECTORS

Extended Only

Scalars & Vectors

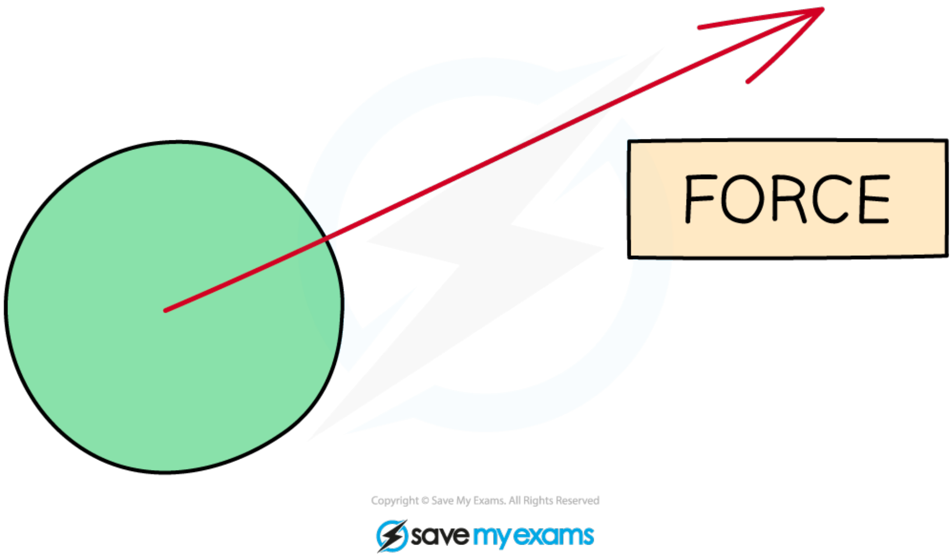
- Quantities can be one of two types: A scalar or a vector
- **Scalars are quantities that have only a magnitude** (a number describing how big they are)
- **Vectors have both magnitude and direction**

The cars in the above diagram have the same speed (a scalar quantity) but different velocities (a vector quantity)

1. General Physics

YOUR NOTES

- **Force is a vector quantity** – it has both magnitude and direction



The force is represented by the arrow. Its length gives the magnitude (size) of the force and the arrow also shows its direction

1. General Physics

YOUR NOTES

- Some **common scalars and vectors** are given below

SCALAR	VECTOR
DISTANCE	DISPLACEMENT
SPEED	VELOCITY
TIME	ACCELERATION
ENERGY	FORCE
MASS	WEIGHT
	MOMENTUM

- **Note:** Some vector quantities (such as displacement and velocity) are very similar to some corresponding scalar quantities (distance and speed)

1. General Physics

YOUR NOTES
⬇

Extended Only

Adding Vectors

- Vectors can be added together to produce a resultant vector. The rules for doing this, however, are slightly different to scalars:
  - If two vectors point in the same direction, the resultant vector will also have the same directions and its value will be the result of adding the magnitudes of the two original vectors together
  - If two vectors point in opposite directions then subtract the magnitude of one of the vectors from the other one. The direction of the resultant will be the same as the larger of the two original vectors

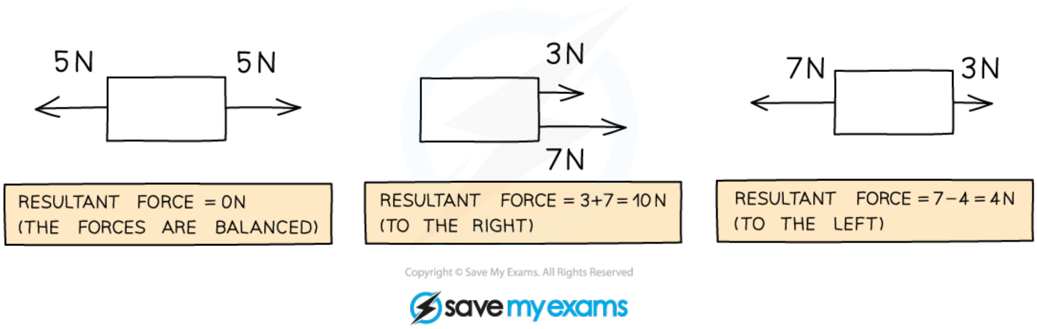


Diagram showing the result of adding two aligned vectors (forces) together

1. General Physics

YOUR NOTES

- If the two vectors point in completely different directions, then the value of the resultant vector can be found graphically:
  - Draw an arrow representing the first vector
  - Now starting at the head of the first arrow, draw a second arrow representing the second vector
  - The resultant vector can be found by drawing an arrow going from the tail of the first vector to the tip of the second vector

Diagram showing an example of the "tip-to-tail" addition of two vectors

Exam Question: Medium

A skydiver is falling at terminal velocity. He has not yet opened his parachute.

He opens his parachute.

What is the direction of both the velocity of the skydiver, and his acceleration?

	Direction of the skydiver's velocity	Direction of the skydiver's acceleration
A	upwards	upwards
B	upwards	downwards
C	downwards	upwards
D	downwards	downwards

1. General Physics

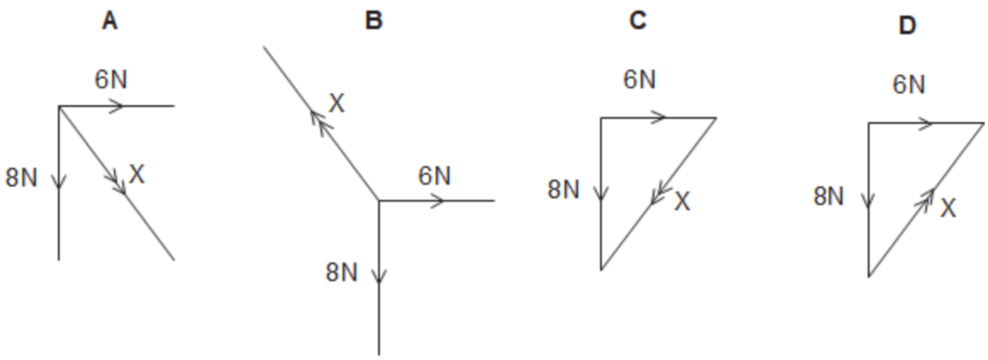
YOUR NOTES
⬇

Exam Question: Hard

Two forces act on an object at right angles to each other: a 6 N force and an 8 N force.

The diagrams below show the two forces.

Which diagram also shows the correct resultant force?



> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

YOUR NOTES

1.6 MOMENTUM

1.6.1 MOMENTUM

Extended Only

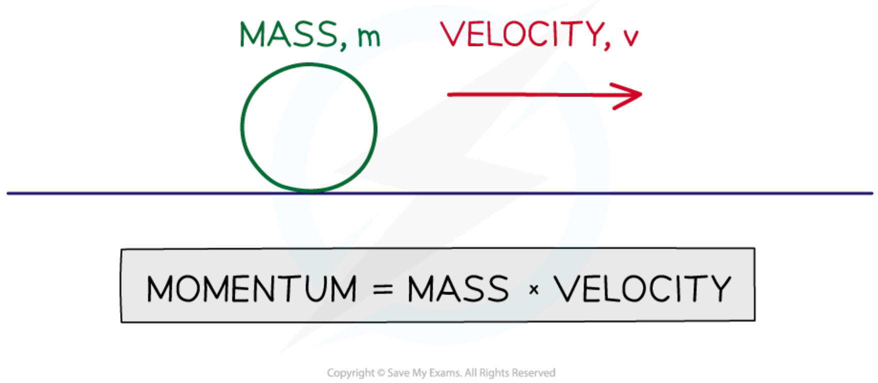
Momentum Defined

- Momentum is defined by the equation:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

$$p = m \times v$$

(where *p* stands for momentum)

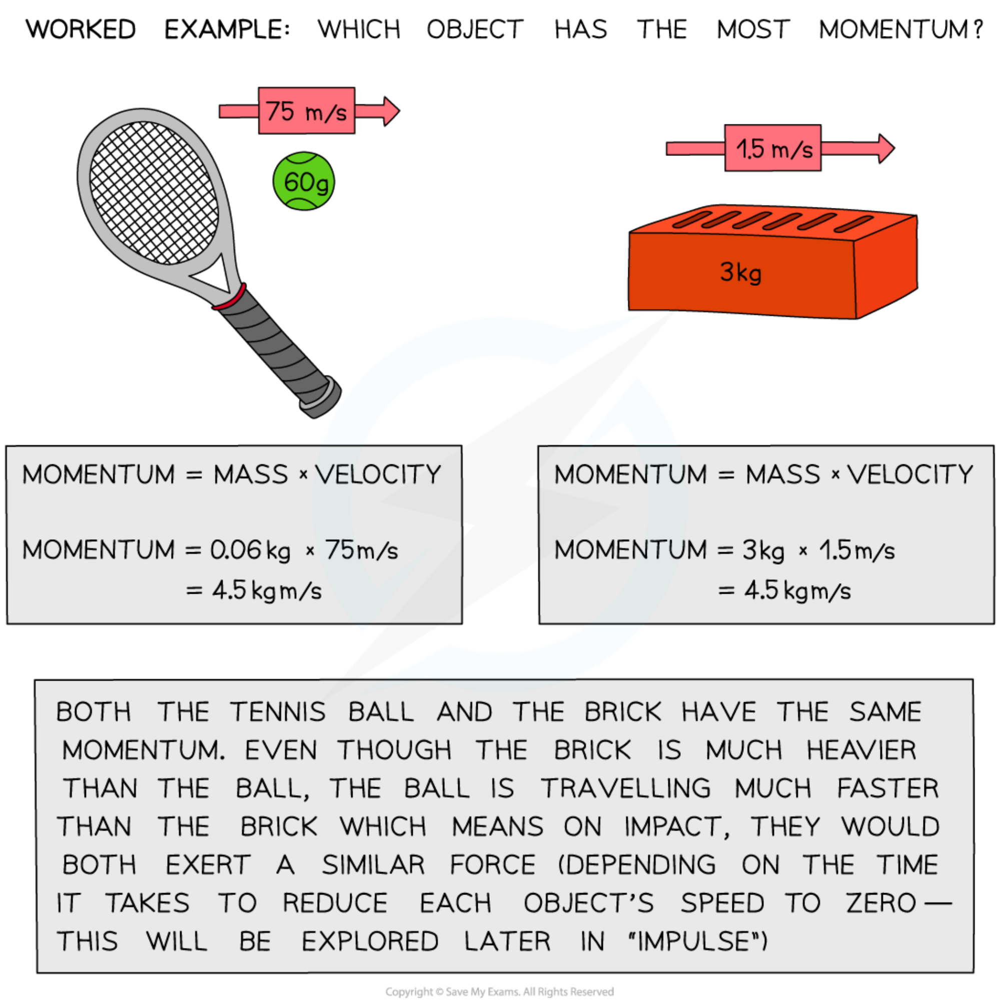


Momentum is defined as the product of mass and velocity

1. General Physics

YOUR NOTES
⬇

- **The units of momentum are kg m/s** (the units of mass multiplied by the units of velocity)
- **Momentum is a vector quantity** – it has direction as well as magnitude
- This means that momentum can be **negative as well as positive**:
    - If an object travelling to the right has positive momentum, an object travelling in the opposite direction (to the left) will have negative momentum



1. General Physics

YOUR NOTES

Extended Only

The Conservation of Momentum

- **In the absence of external forces (such as friction), the total momentum of a system remains the same**
- This means that in a collision, the sum of the momentums before the collision will be the same as the sum of momentums after the collision

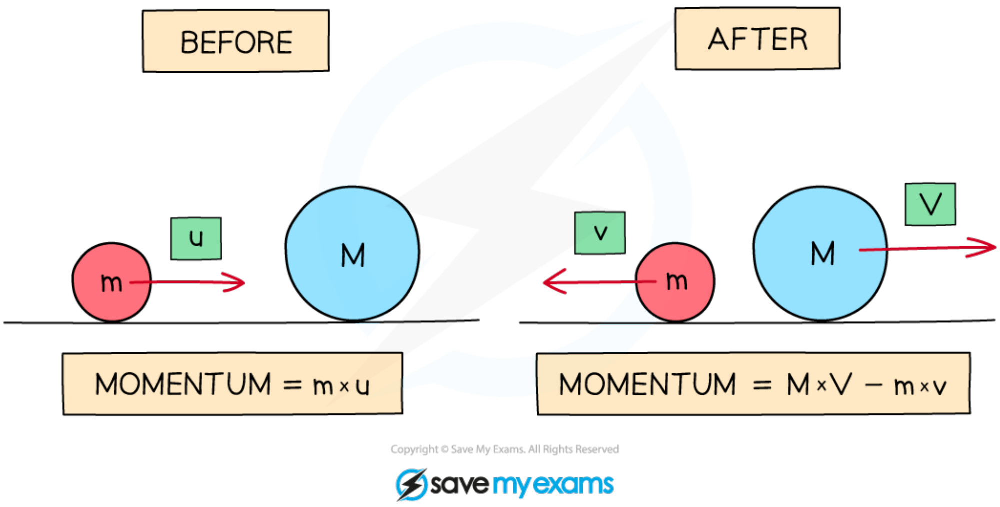


Diagram showing the total momentum of a system before and after a collision

- In the above diagram the total momentum before and the total momentum after must be equal:

$$m \times u = M \times V - m \times v$$

- Note that because the red ball is travelling to the left after the collision, its momentum will be negative – hence the minus sign in the above equation

1. General Physics

YOUR NOTES

Extended Only

Impulse

- When a resultant (unbalanced) force acts on a mass, the momentum of that mass will change
- The **impulse of a force** is equal to that force multiplied by the time for which it acts:

impulse = F × t

- The change in momentum of a mass is equal to the impulse provided by the force:

impulse = change in momentum

F × t = mv - mu

(Where *u* is the initial velocity of the mass and *v* is the final velocity of the mass)

Exam Question: Easy

Which of these is the correct definition for momentum and its unit?

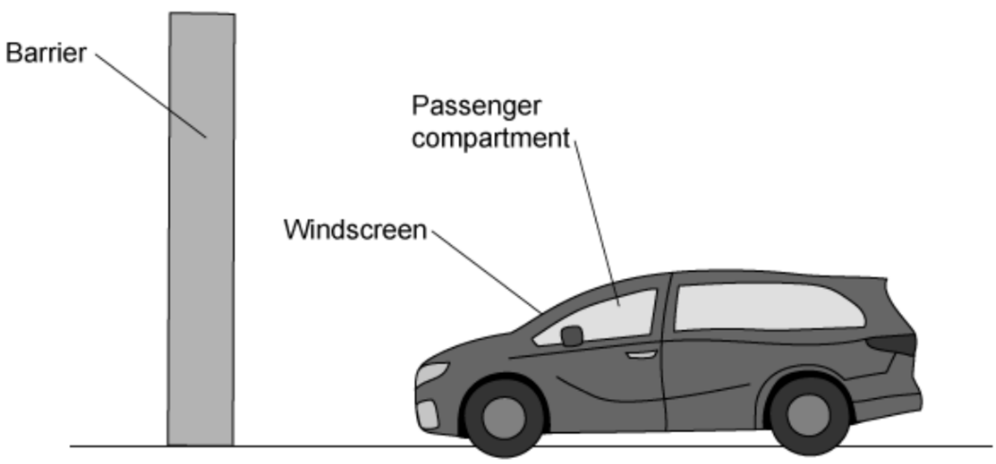
	definition	unit
A	p = m ÷ v	N/m
B	v = p × m	kgm/s
C	v = p ÷ m	kgm/s
D	p = m × v	N/kg

1. General Physics

YOUR NOTES

Exam Question: Medium

A passenger of mass 90 kg is involved in a minor car crash.



The car approaches a solid barrier at 32 m/s. It crashes into the barrer and stops in 0.2s.

Determine the impulse that must be applied to the car to bring it to rest.

A 2.8 Ns

B 14 Ns

C 580 Ns

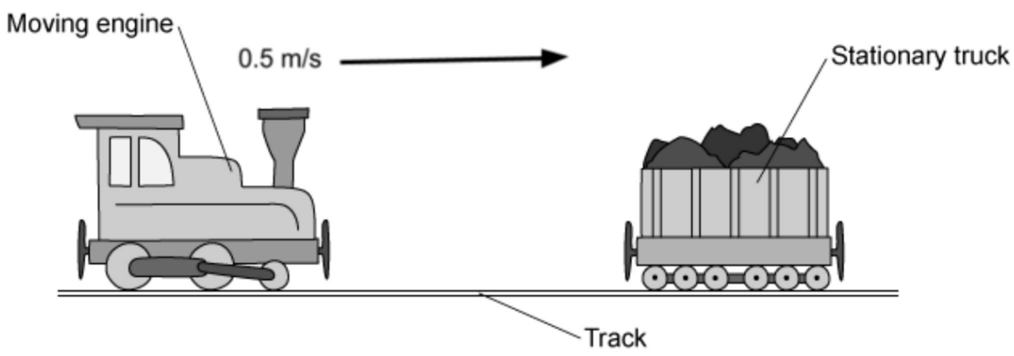
D 2900 Ns

1. General Physics

YOUR NOTES

Exam Question: Hard

A toy train is rolling at a constant speed on a level track. The train collides with a stationary toy truck and joins with it.



Before the collision, the toy train is travelling at 0.5 m/s. The train and the stationary truck both have a mass of 75 g.

Calculate the momentum of the toy train before the collision.

A 37.5 kg m/s

B 0.04 kg m/s

C 0.38 kg m/s

D 0.08 kg m/s

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

YOUR NOTES
⬇

1.7 ENERGY, WORK & POWER

1.7.1 ENERGY

The Conservation of Energy

- **Energy is the capacity of something to do work:**
  - If something contains a store of energy it is able to do work
  - If something does not store energy then it will not work
- The law of conservation of energy states that:
  - **Energy cannot be created or destroyed, it can only change from one form to another**
- What this means is that the total amount of energy in a closed system remains constant, although how much of each form there is may change

Some examples:

- A falling object (in a vacuum): Gravitational potential energy → Kinetic energy
- A gas cooker: Chemical → Internal (Heat)
- An LED (Light Emitting Diode): Electrical → Light

1. General Physics

YOUR NOTES

Extended Only

Conservation of Energy in Multiple Stages

- Many processes involve several steps before energy ends up in its final form

For example:

- A fossil fuel power station takes chemical energy (the fuel) and uses it to produce electrical energy, but the individual steps are:

```
Chemical → Internal (heat) → Kinetic (steam) → Kinetic (turbine and generator) → Electrical
```

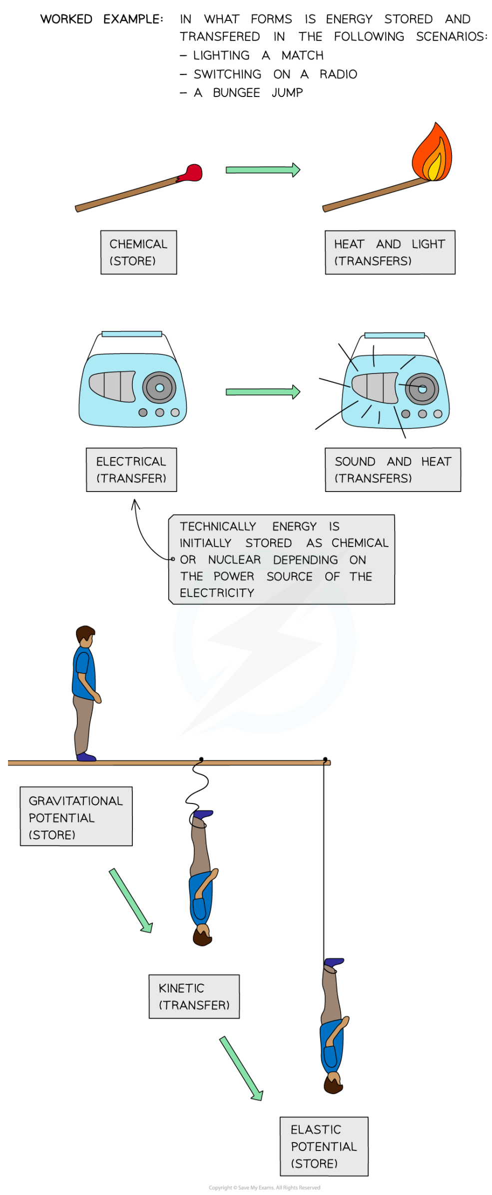
Types of Energy

- Energy can exist in a number of forms
  For your IGCSE examination you are expected to know about the following:

FORM	WHAT IS IT?
KINETIC	THE ENERGY OF A MOVING OBJECT.
GRAVITATIONAL POTENTIAL	THE ENERGY SOMETHING GAINS WHEN YOU LIFT IT UP, AND WHICH IT LOSES WHEN IT FALLS.
ELASTIC	THE ENERGY OF A STRETCHED SPRING OR ELASTIC BAND.(SOMETIMES CALLED STRAIN ENERGY)
CHEMICAL	THE ENERGY CONTAINED IN A CHEMICAL SUBSTANCE.
NUCLEAR	THE ENERGY CONTAINED WITHIN THE NUCLEUS OF AN ATOM.
INTERNAL	THE ENERGY SOMETHING HAS DUE TO ITS TEMPERATURE (OR STATE). (SOMETIMES REFERRED TO AS THERMAL OR HEAT ENERGY)

1. General Physics

YOUR NOTES



Worked example showing how energy is stored and transferred

1. General Physics

YOUR NOTES

Energy Transfer

- In addition to the six forms of energy mentioned above, there are four ways in which energy can be transferred from one form to another:

TRANSFER	WHAT IS IT?
FORCES (MECHANICAL WORKING)	WHEN A FORCE ACTS ON A BODY, ENERGY CAN BE TRANSFERRED BETWEEN TWO FORMS.
ELECTRICAL CURRENTS (ELECTRICAL WORKING)	ELECTRICITY CAN TRANSFER ENERGY FROM A POWER SOURCE, DELIVERING IT TO COMPONENTS WITHIN A CIRCUIT.
HEATING	INTERNAL ENERGY CAN BE TRANSFERRED FROM PLACE TO PLACE BY THE PROCESSES OF CONDUCTION, CONVECTION AND THERMAL RADIATION.
WAVES	LIGHT AND SOUND CARRY ENERGY AND SO CAN TRANSFER IT BETWEEN PLACES.

1. General Physics

YOUR NOTES

Extended Only

Energy Dissipation

- When energy is transferred from one form to another, **not all of the energy will end up in the desired form** (or place)
- This lost energy often ends up being **dissipated** (spreading out into the environment), usually in the form of **heat, light or sound**

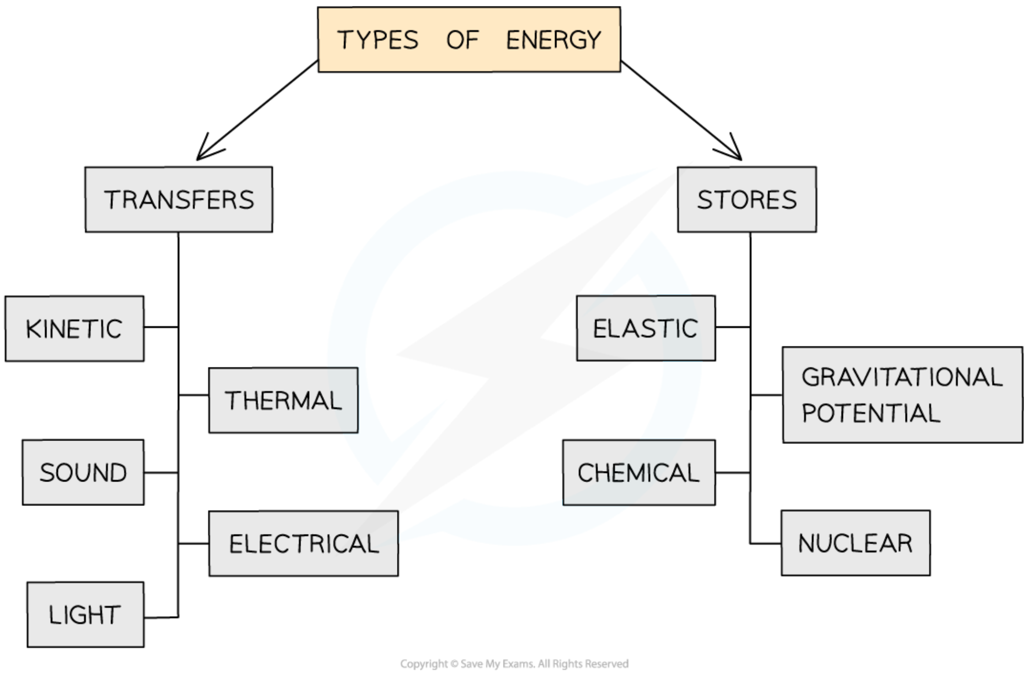


Diagram showing the forms of energy transfers and stores

1. General Physics

YOUR NOTES
⬇

Exam Tip

The classification of energy into forms and transfers, as mentioned above, is a fairly new approach and so there is a chance that you may have been taught this topic slightly differently or have come across different approaches elsewhere.

Don’t worry if this is the case, but you will need to become familiar with the above classification as there is a very good chance that it will be used in your examinations.

1. General Physics

YOUR NOTES
⬇

1.7.2 KE & GPE

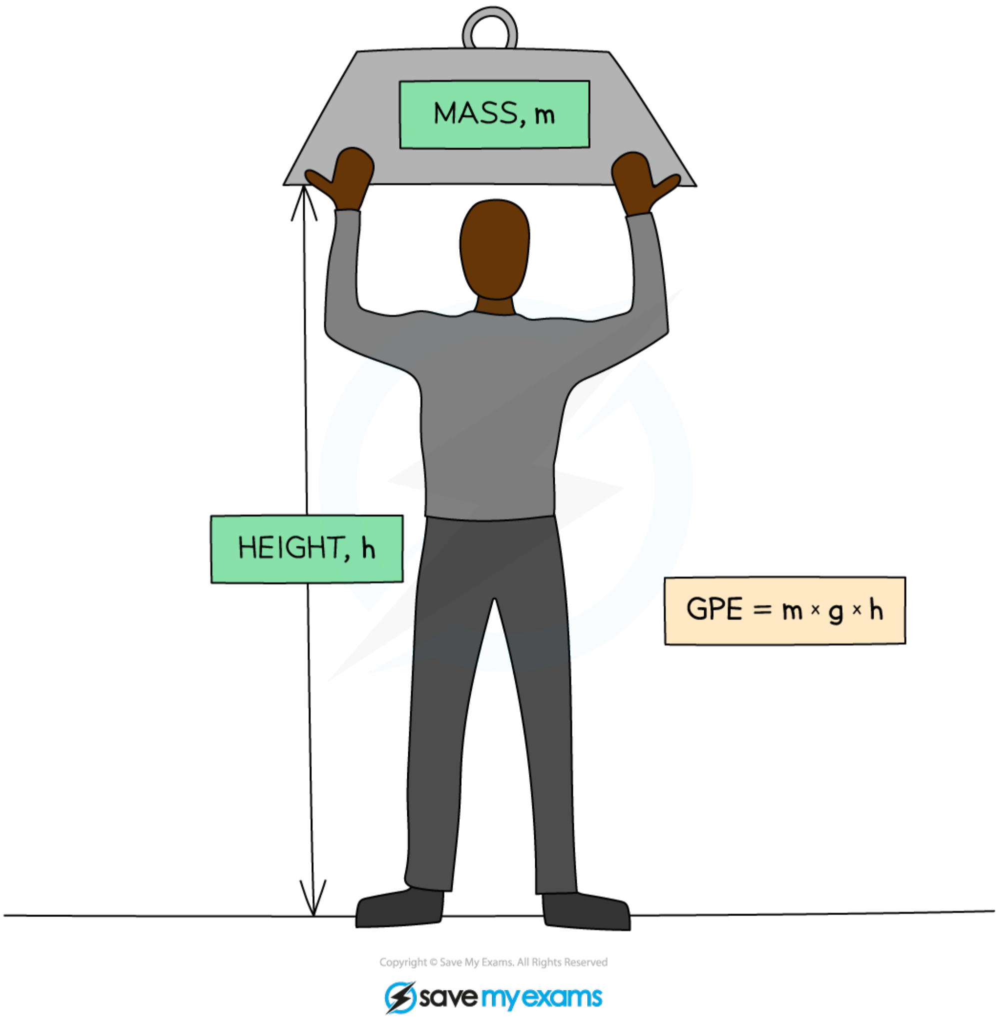
1. General Physics

YOUR NOTES

Extended Only

Gravitational Potential Energy

- The **gravitational potential energy (GPE)** of an object is the energy it has due to its height in a gravitational field:
  - If an object is lifted up it will gain GPE
  - If it falls, it will lose GPE



Gravitational potential energy: The energy an object has when it is lifted up

1. General Physics

YOUR NOTES
⬇

- The GPE of an object is related to its mass (*m*), height (*h*) and the gravitational field strength (*g*):

$$\text{GRAVITATIONAL POTENTIAL ENERGY} = \text{MASS} \times \text{GRAVITATIONAL FIELD STRENGTH} \times \text{HEIGHT}$$

$$GPE = m \times g \times h$$

- The gravitational field strength (*g*) on the Earth is approximately **10 N/kg** (You will always be told this value in your examination paper)

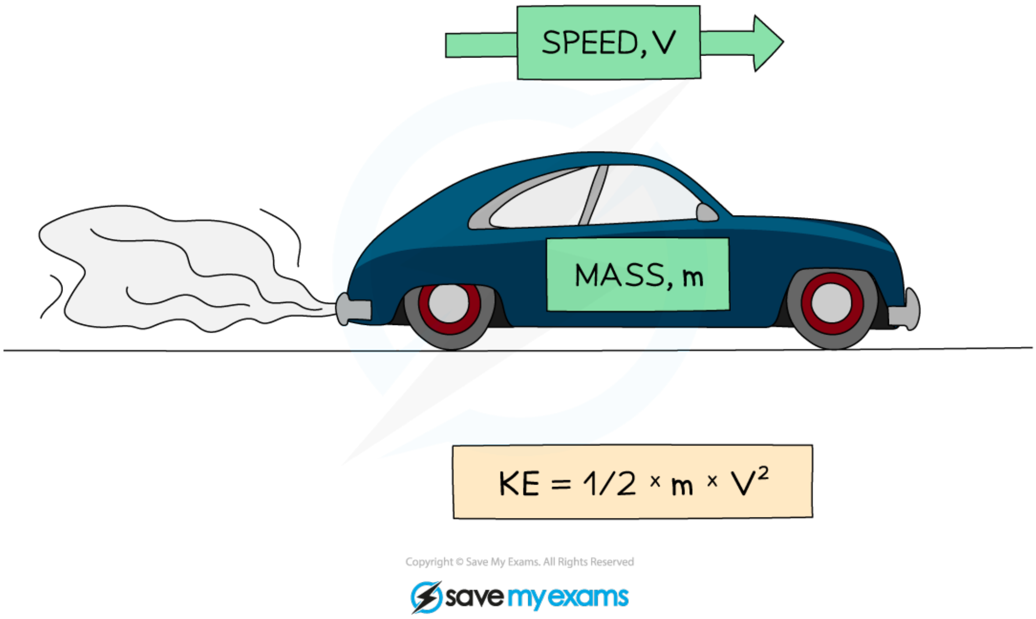
1. General Physics

YOUR NOTES

Extended Only

Kinetic Energy

- The **kinetic energy (KE)** of an object is the energy it has as a result of its speed



Kinetic energy: the energy an object has when it is moving

- It is related to the mass (m) and speed (v) of the object by the equation:

$$\text{KINETIC ENERGY} = \frac{1}{2} \times \text{MASS} \times \text{SPEED}^2$$

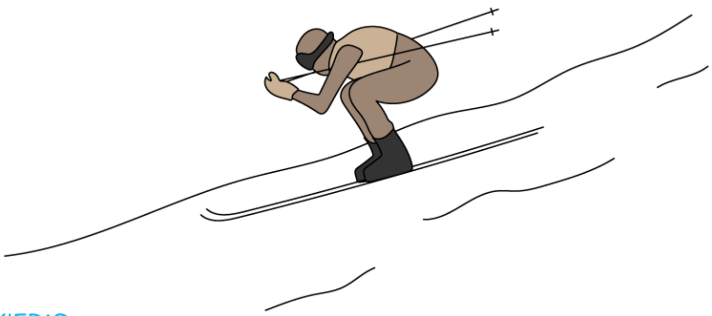
$$KE = \frac{1}{2} \times m \times v^2$$

(**Note**: In the above equation only the speed is squared – not the mass or the ½)

1. General Physics

YOUR NOTES
⬇

WORKED EXAMPLE: A SKIER TAKES PART IN A DOWNHILL RACE. THE MASS OF THE SKIER, INCLUDING HIS EQUIPMENT, IS 75kg. FROM START TO FINISH, THE TOTAL VERTICAL CHANGE IN HEIGHT IS 880m.



WHAT IS THE SKIER'S

– GRAVITATIONAL POTENTIAL ENERGY AT THE START OF THE RACE?

$GPE = mgh$

$GPE = 75 \times 9.8 \times 880 = 646{,}800\,J$

$GPE = 6.5 \times 10^5\,J$

– KINETIC ENERGY AT THE END OF THE RACE?

$KE = 6.5 \times 10^5\,J$

ALL OF THE SKIER'S GRAVITATIONAL POTENTIAL ENERGY AT THE TOP OF THE SLOPE WILL BE TRANSFERED TO KINETIC ENERGY AT THE BOTTOM (ASSUMING NONE IS DISSIPATED AS HEAT AND SOUND)

Worked example showing how GPE transfers to KE

1. General Physics

YOUR NOTES
⬇

1.7.3 ENERGY RESOURCES

Descriptions & Forms

- Energy resources are large banks of energy that can be transferred into a form that can be used by society, such as electrical energy
- The table below gives a brief description of the main energy resources, along with the form of energy contained in that resource

ENERGY RESOURCE	DESCRIPTION	ENERGY FORM
FUELS	BURNING FUELS PRODUCE HEAT, WHICH CAN BE USED TO TURN WATER INTO STEAM TO TURN TURBINES	CHEMICAL
WATER	HYDROELECTRIC AND TIDAL POWER USE THE GPE OF WATER TO TURN TURBINES WHICH GENERATE ELECTRICITY	GPE
WAVES	WAVES ALSO CONTAIN ENERGY, IN THE FORM OF KE	KE
GEOTHERMAL	HEAT FROM UNDERGROUND CAN BE USED TO CREATE STEAM, WHICH SPINS TURBINES PRODUCING ELECTRICITY.	INTERNAL
NUCLEAR FISSION	NUCLEAR FUEL IS REACTED, PRODUCING HEAT WHICH CREATES STEAM	NUCLEAR
SOLAR HEATING	LIGHT FROM THE SUN CAN BE USED TO WARM WATER PASSING THROUGH BLACK PIPES	LIGHT
SOLAR CELLS	PHOTOVOLTAIC CELLS CAN USE LIGHT TO CREATE ELECTRICITY	LIGHT
WIND	WIND TURBINES CAN BE USED TO PRODUCE ELECTRICITY	KE

Advantages and Disadvantages

1. General Physics

YOUR NOTES

- All energy resources have advantages and disadvantages associated with them

ENERGY RESOURCE	RENEWABLE?	ADVANTAGES	DISADVANTAGES
FOSSIL FUELS	NO	RELIABLE. CAN PRODUCE ENERGY ON A LARGE SCALE. COST–EFFECTIVE	PRODUCES SIGNIFICANT GREENHOUSE GASES AND POLLUTION.
WATER	YES	HYDRO IS RELIABLE AND CAN PRODUCE LARGE AMOUNT OF ENERGY AT SHORT NOTICE. PRODUCES NO POLLUTION OF GREENHOUSE GASES.	TIDAL IS NOT RELIABLE (BUT IS PREDICTABLE). CAN INVOLVE FLOODING LARGE AREA, DESTROYING IMPORTANT WILDLIFE HABITATS. VERY FEW SUITABLE SITES. THE DAMS CAN BE VERY EXPENSIVE TO BUILD, BUT RUNNING COSTS ARE LESS THAN FOR FOSSIL FUELS.
GEOTHERMAL	YES	RELIABLE. CAN BE COST–EFFECTIVE (IF BUILT IN THE RIGHT PLACES).	CAN RESULT IN THE RELEASE OF HARMFUL GASES FROM UNDERGROUND. NOT MANY PLACES ARE SUITABLE. DOESN'T PRODUCE ENERGY ON A LARGE SCALE.
NUCLEAR	NO	RELIABLE AND CAN PRODUCE ENERGY ON VERY LARGE SCALES. PRODUCES NO GREENHOUSE GASES OR POLLUTION. A LARGE AMOUNT OF ENERGY IS PRODUCED FROM A SMALL AMOUNT OF FUEL.	PRODUCES DANGEROUS RADIOACTIVE WASTE THAT CAN TAKE THOUSANDS OF YEARS TO DECAY.MPOWER STATIONS ARE EXPENSIVE TO BUILD AND VERY EXPENSIVE TO DECOMMISSION.
SOLAR	YES	PRODUCES NO GREENHOUSE GASES OR POLLUTION. GOOD FOR PRODUCING ENERGY IN REMOTE PLACES.	NOT RELIABLE (ONLY WORKS WHEN SUNNY). DOESN'T PRODUCE ENERGY ON A LARGE SCALE. PANELS ARE VERY EXPENSIVE TO BUY AND THE COST–EFFECTIVENESS IS QUESTIONABLE WITHOUT GOVERNMENT SUBSIDIES.
WIND	YES	PRODUCES NO GREENHOUSE GASES OR POLLUTION. LAND CAN STILL BE USED FOR FARMING.	NOT RELIABLE. CAN BE NOISY AND UGLY. NOT EVERYWHERE IS SUITABLE. TURBINES CAN BE EXPENSIVE TO BUILD (RELATIVE TO THE ENERGY THEY PRODUCE), BUT RUNNING COSTS ARE LOW.

1. General Physics

YOUR NOTES
⬇

- Some points to note:
  - A **renewable energy** resource is one that is replenished at a faster rate than the rate at which it is being used
    As a result of this, renewable energy resources cannot run out
  - A **reliable energy** resource is one that can produce energy at any time
    Non-reliable resources can only produce energy some of the time (e.g. when it's windy)

1. General Physics

YOUR NOTES

Extended Only

The Sun

- Most of our energy resources on the Earth come from the Sun:
    - The Sun heats up the atmosphere, creating wind and producing waves
    - Water evaporated by the Sun falls as rain, filling up reservoirs
    - Plants grown using sunlight form the basis for fuels – both biofuels and fossil fuels
- The Sun's energy is produced by through the process of nuclear fusion in its core
    - Nuclear fusion involves the collision (and bonding) of hydrogen nuclei to form helium nuclei, releasing nuclear energy in the process
- Some forms of energy, however, do not come from the Sun
  These include:
    - Geothermal – this comes from heat produced in the Earth's core
    - Nuclear – this comes from elements which make up a small proportion of the Earth's crust
    - Tidal – this comes (mainly) from the gravitational attraction of the moon

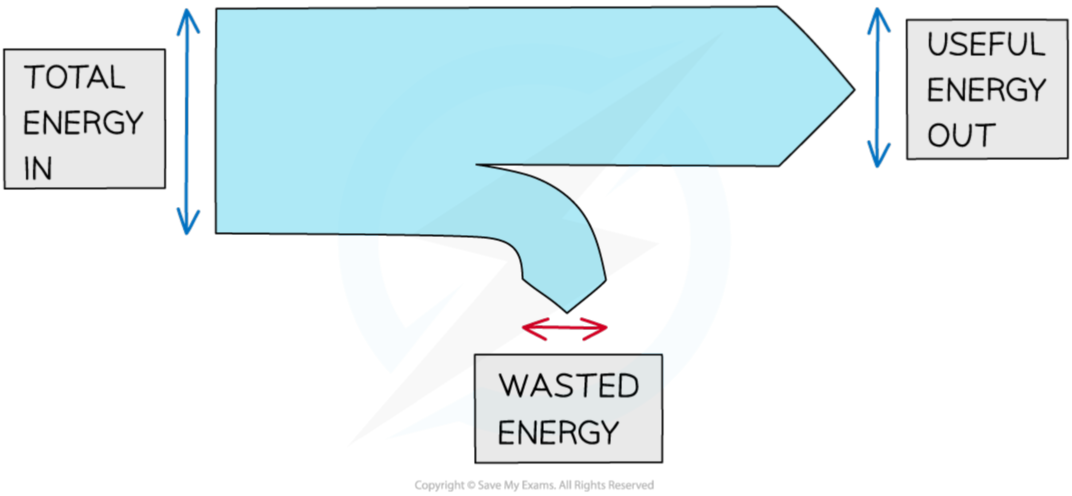
1. General Physics

YOUR NOTES ⬇

1.7.4 EFFICIENCY

Efficient & Inefficient Systems

- Whenever energy is transferred from one form to another, **some of that energy is usually wasted** and is transferred away from the system, usually in the form of heat or waves (light and sound)

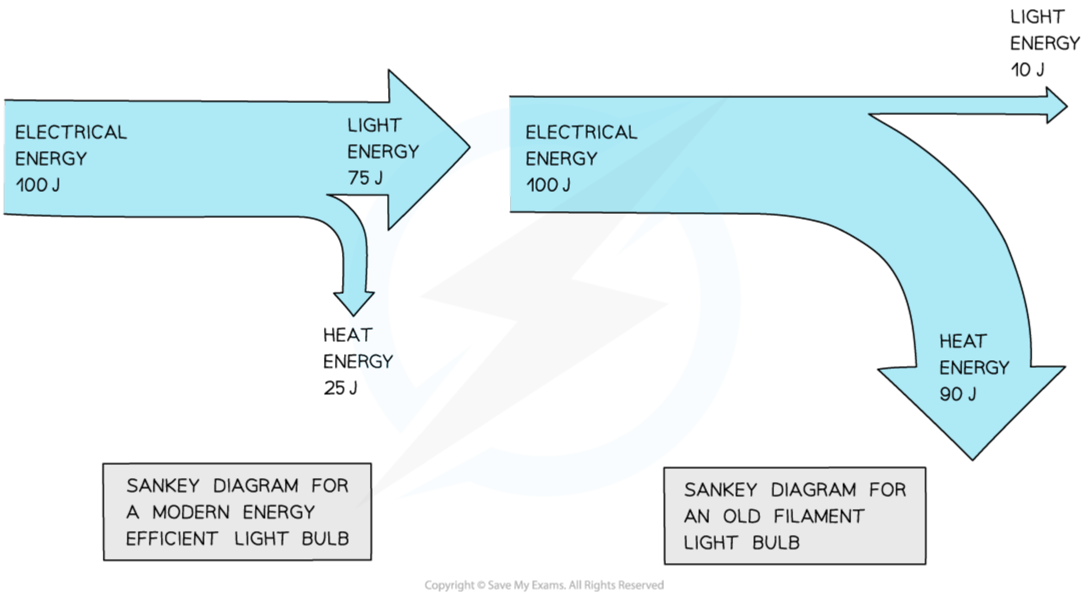


Whenever energy is transformed, some of the original energy usually ends up in an unwanted (wasted) form

1. General Physics

YOUR NOTES
⬇

- An efficient system is one where most of the energy going into that system ends up in the form that is wanted
  Eg. An LED light bulb is efficient because most of the electrical energy (90%) ends up as light
- An inefficient system is one where most of the energy ends up in forms that weren't wanted
  Eg. An old incandescent light bulb is inefficient because only a small amount of the electrical energy (5%) ends up as light



Sankey diagrams comparing modern and old light bulbs

1. General Physics

YOUR NOTES

Extended Only

Calculating Efficiency

- The efficiency of a system is **the percentage of energy transferred from the original store that ends up in the intended form**
- Efficiency can be calculated using the following equation:

$$\text{EFFICIENCY} = \frac{\text{USEFUL ENERGY OUTPUT}}{\text{TOTAL ENERGY INPUT}} \times 100\%$$

- Efficiency can also be written in terms of power) the energy per second):

$$\text{EFFICIENCY} = \frac{\text{USEFUL POWER OUTPUT}}{\text{TOTAL POWER INPUT}} \times 100\%$$

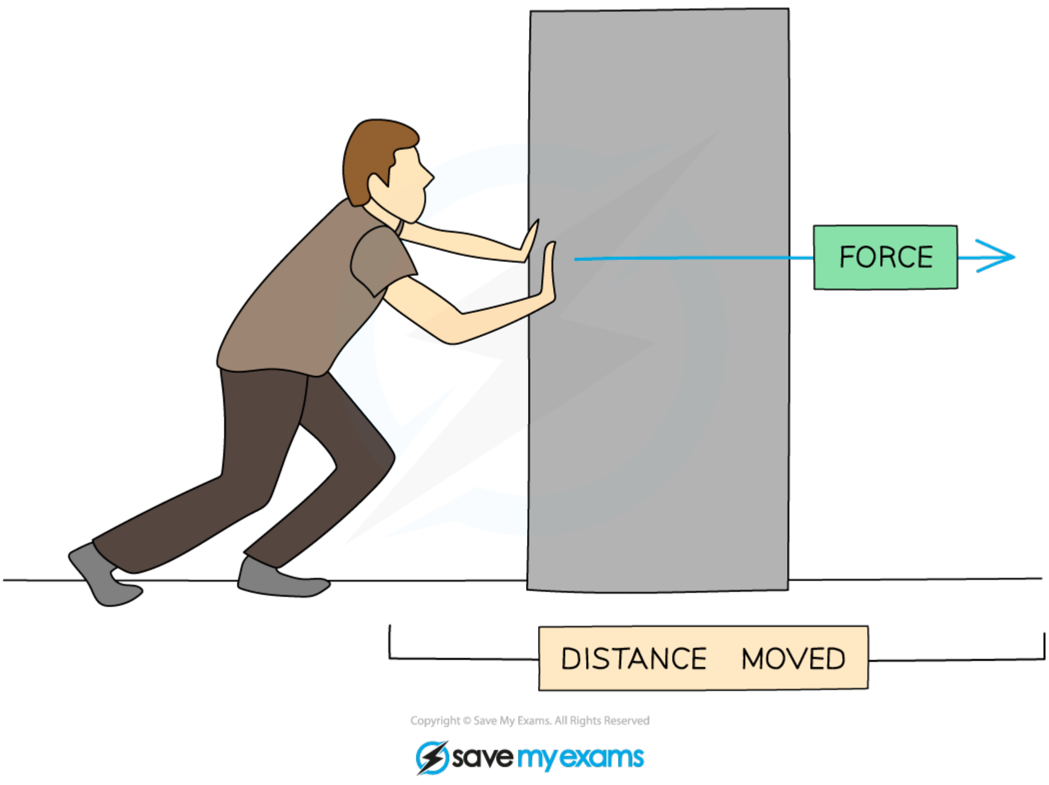
1. General Physics

YOUR NOTES

1.7.5 WORK

Work Done

- **Energy is the capacity of something to do work**
- **Work is done whenever a force acts on an object that moves** (or is moving) in the direction of the force
  - The greater the force, the greater the work
  - The larger the distance moved, the larger the work



Work is done when a force is used to move an object a distance

1. General Physics

YOUR NOTES
⬇

- Whenever any work is done, energy gets transferred (mechanically) from one form to another
- The amount of energy transferred (in joules) is equal to the work done (also in joules)

energy transferred (J) = work done (J)

- Usually, if a force acts in the direction that an object is moving then the object will gain energy
- If the force acts in the opposite direction to the movement then the object will lose energy

1. General Physics

YOUR NOTES

Extended Only

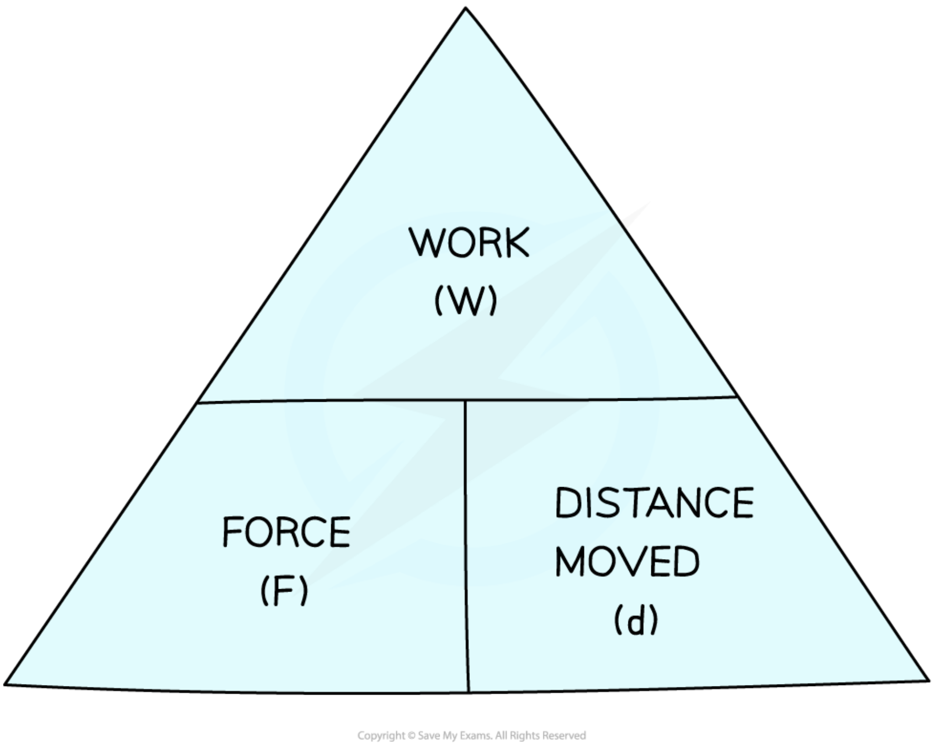
Calculating Work Done

- The amount of work that is done if related to the size of the force and the distance moved by the object in the direction of the force:

```
work done  =  force × distance moved
```

```
W = F× d
```

- You can rearrange this equation with the help of the formula triangle:



Use the formula triangle to help you rearrange the equation

- **The units of work are joules (J)** (the same as the units of energy), but can also be given as newton metres (Nm)

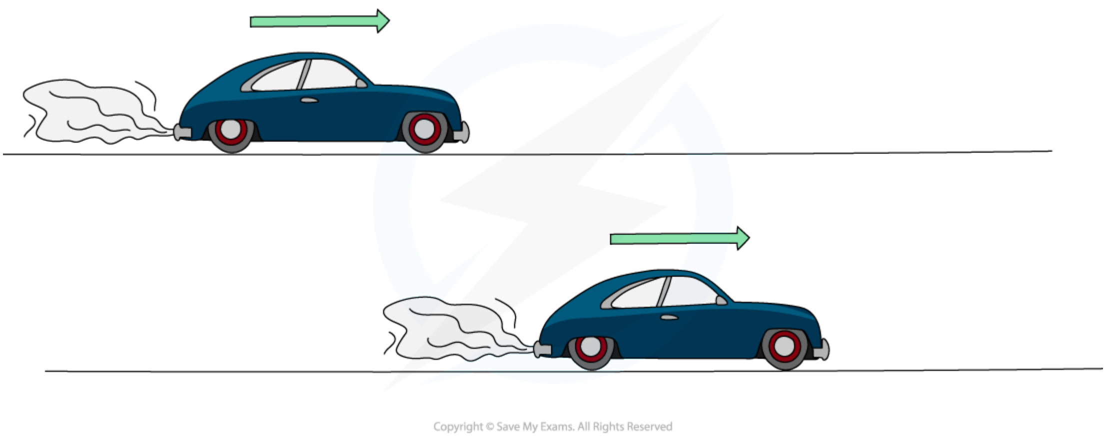
1. General Physics

YOUR NOTES
⬇

1.7.6 POWER

Power, Work Done & Time Taken

- Machines, such as car engines, transfer energy from one form to another every second
- **The power of a machine is the rate at which the machine transfers energy**
    - The greater the rate at which energy is transferred, the greater the power



Two identical cars accelerating to the same final speed will both gain the same amount of energy. But if one of them reaches that speed sooner, it will have a greater power

- Because work done is equal to energy transferred, the power is also equal to the rate of doing work

1. General Physics

YOUR NOTES

Extended Only

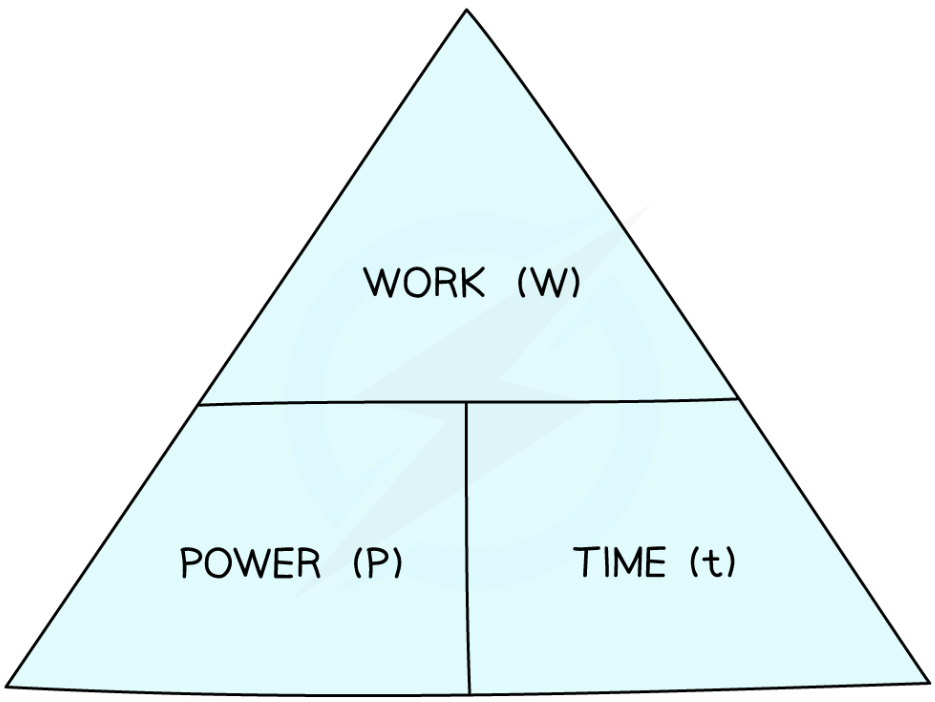
Calculating Power

- **Power is the amount of energy transferred (or work done) every second.**
- Power, work and time are related by the following equation:

$$\text{POWER} = \frac{\text{ENERGY TRANSFERRED}}{\text{TIME TAKEN}}$$

$$P = \frac{E}{t}$$

- You can rearrange this equation with the help of the formula triangle:



1. General Physics

YOUR NOTES

Use the formula triangle to help you rearrange the equation

- **The unit of power is the watt (W)**, which is the same as a joule per second (J/s)

Exam Tip

Think of power as “energy per second”. Think of it this way will help you to remember the relationship between power and energy
“Watt is the unit of power?”

Exam Question: Easy

Which energy resource does not use the Sun as the source of its energy?

A Hydroelectric

B Wind

C Nuclear

D Coal

1. General Physics

YOUR NOTES

Exam Question: Hard

A large truck of mass 5000 kg is travelling at 5.0 m/s.

A motorbike has a mass of 200 kg. Both the truck and the motorbike have the same kinetic energy.

What is the speed of the motorbike?

A 25 m/s

B 10 m/s

C 125 m/s

D 2.5 m/s

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK

1. General Physics

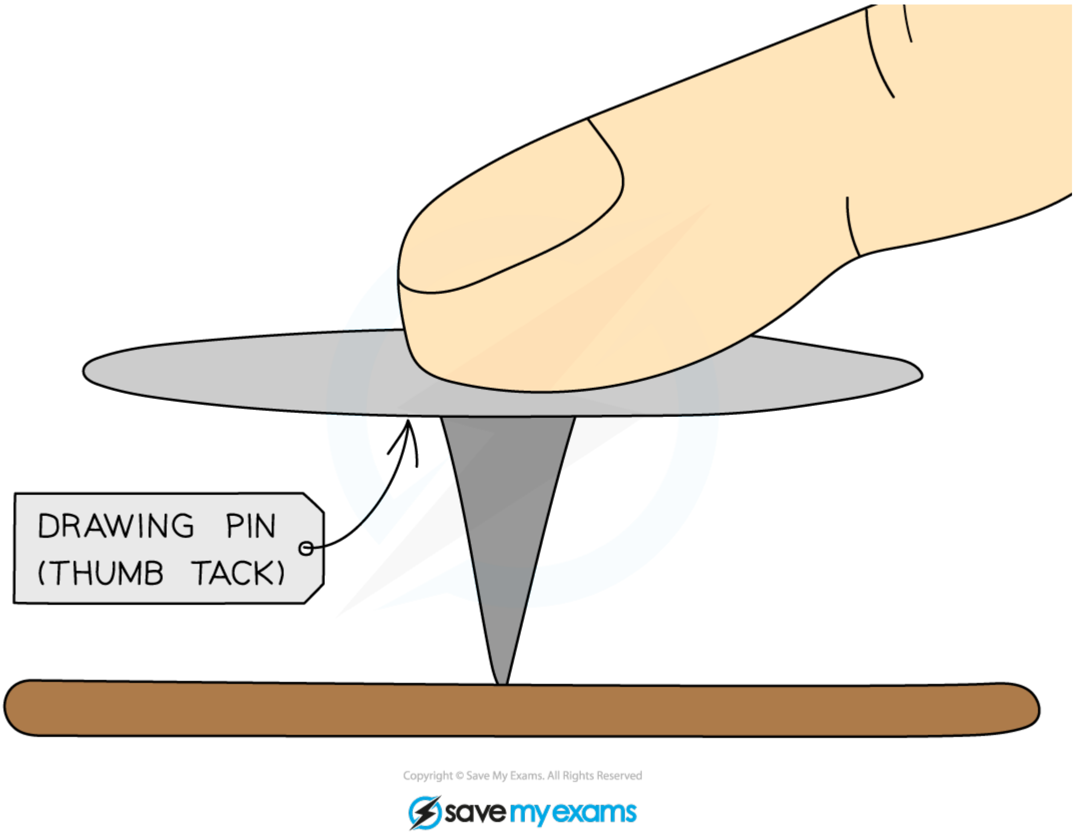
YOUR NOTES
⬇

1.8 PRESSURE

1.8.1 PRESSURE

Pressure, Force & Area

- **Pressure is the concentration of a force**:
  - If a force is spread over a large area it will result in a small pressure
  - If it is spread over a small area it will result in a large pressure



When you push a drawing pin, it goes into the surface (rather than your finger) because the force on the surface is more concentrated

1. General Physics

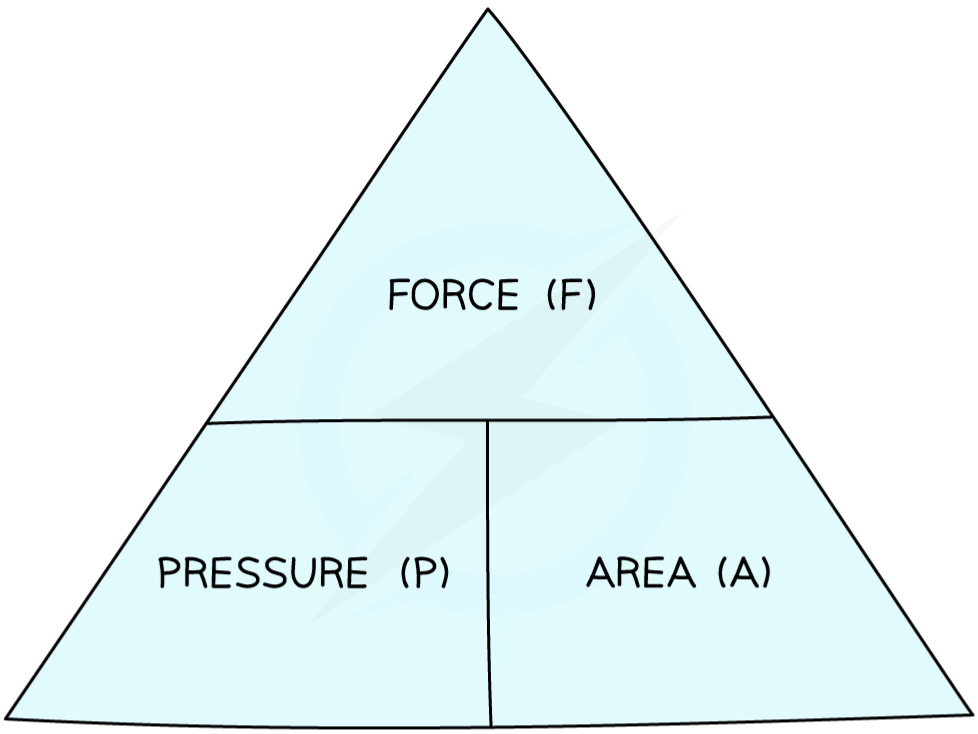
YOUR NOTES

- Pressure is related to force and area by the equation:

$$\text{PRESSURE} = \frac{\text{FORCE}}{\text{AREA}}$$

$$P = \frac{F}{A}$$

- You can rearrange this equation with the help of the formula triangle:



Use the formula triangle to help you rearrange the equation

1. General Physics

YOUR NOTES

- The units of pressure depend on the units of area:
  - If the area is measured in **cm^2** (and the force in **N**), then the pressure will be in **N/cm^2**
  - If the area is measured in **m^2** (and the force in **N**), then the pressure will be in **N/m^2**
- Pressure can also be measured in pascals, Pa

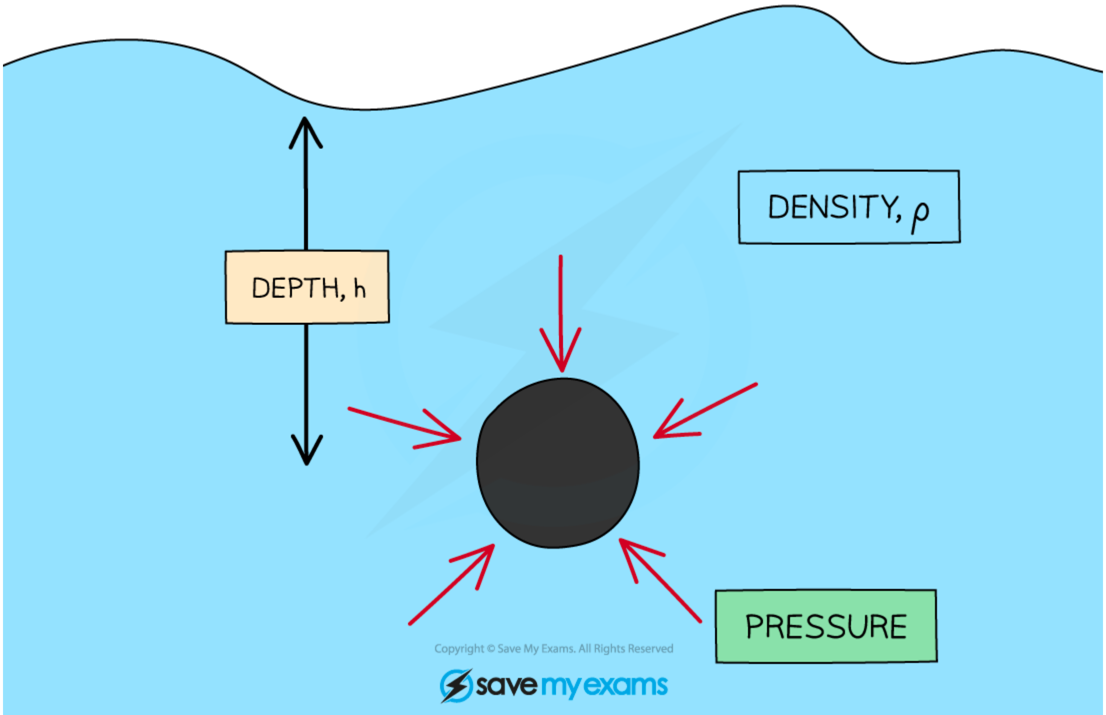
  1 Pa is the same as 1 N/m^2

1. General Physics

YOUR NOTES
⬇

Pressure in Liquids: Basics

- When an object is immersed in a liquid, the liquid will exert a pressure, squeezing the object
- This pressure is exerted evenly across the whole surface of the liquid, and in all directions



Pressure, at a point in a liquid, acts equally in all directions

- The pressure in the liquid depends upon a couple of factors:
    - **The depth of the liquid**
    - **The density of the liquid**
- The bigger either of these factors, the greater the pressure

1. General Physics

YOUR NOTES
⬇

Extended Only

Pressure in Liquids

- The size of this pressure depends upon the density (***ρ***) of the liquid, the depth (***h***) of the object and the gravitational field strength (***g***):

PRESSURE = HEIGHT × DENSITY × GRAVITATIONAL FIELD STRENGTH

$$P = h \times \rho \times g$$

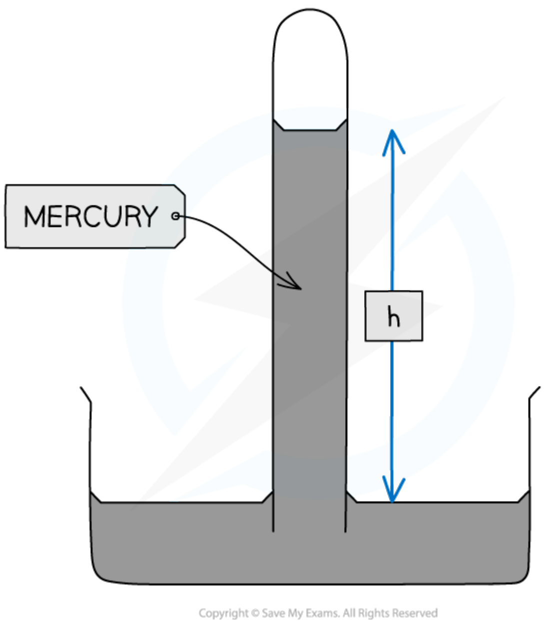
- In the above equation:
  - Pressure is in **pascals** (Pa) – where 1 Pa if the same as 1 N/m^2
  - Depth is in **metres** (m)
  - Density is in **kg/m^3**
- The gravitational field strength on Earth is approximately **10 N/kg**
  (You will always be given this figure)

1. General Physics

YOUR NOTES

Barometers and Manometers

- Barometers and Manometers are devices that measure pressure:
  - A barometer is a device that is used to measure air pressure
  - A manometer is used to measure other pressures
- A simple barometer consists of a column of mercury in an inverted tube, sat in a tray of mercury exposed to the atmosphere



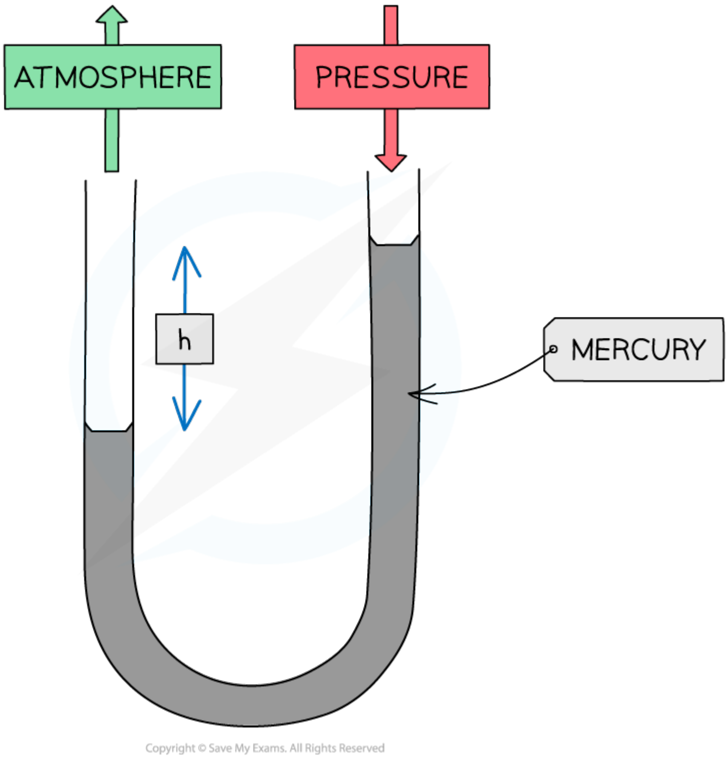
A simple mercury barometer, used to measure atmospheric pressure

- The weight of the mercury in the tube is balanced by atmospheric pressure pushing down on the mercury in the tray
  - If atmospheric pressure increases, a greater length of mercury can be supported in the tube
  - If atmospheric pressure decreases then less mercury will be supported in the tube

1. General Physics

YOUR NOTES

- A simple manometer consists of a U-tube containing mercury



A simple manometer, consisting of mercury in a U-tube

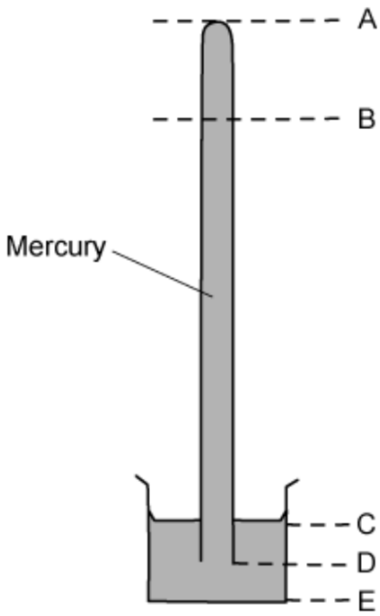
- One end of the U-tube is open to the atmosphere
- The other end is connected to the pressure that is to be measured
- As the measured pressure increases, the mercury is pushed around the U-tube: The greater the pressure, the further it is pushed

1. General Physics

YOUR NOTES

Exam Question: Hard

The diagram shows a mercury barometer, which is used to measure atmospheric pressure.



As the atmospheric pressure decreases, which distance decreases?

A AB

B CD

C DE

D BD

> CHECK YOUR ANSWERS AT SAVEMYEXAMS.CO.UK